Universitat Politècnica de Catalunya

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Research Report

# Downstream Utility of Synthetic Data

## Structural and Inferential Evaluation in Clustering and Regression

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# Abstract

Synthetic data is increasingly used as a privacy-preserving substitute for original data, but its value depends on whether it preserves the conclusions of downstream analyses. This thesis evaluates that analytical utility in two complementary settings: clustering, where utility depends on structural and geometric fidelity, and regression, where utility depends on preserving coefficients, uncertainty, and confidence intervals.

The first study examines whether synthetic data generated with Classification and Regression Trees (CART) preserves the geometry required by clustering workflows. It compares K-Means and Agglomerative Hierarchical Clustering across controlled simulations varying sample size, cluster count, separation, dimensionality, correlation, and distribution family. The results show that CART remains useful for spherical normal settings but introduces block-like artifacts that degrade clustering performance on skewed Gamma data. Across the simulated designs, both clustering methods exhibit broadly similar behavior under synthetic distortion.

The second study evaluates inferential utility for linear and logistic regression. It compares CART-based synthesis with parametric generators using Confidence Interval Overlap, Bias Ratio, and Confidence Interval Width Ratio. Parametric synthesis tends to perform best when its assumptions match the data-generating process, while CART becomes less reliable in high-dimensional, correlated, small-sample, and logistic settings. Overall, the thesis shows that synthetic-data utility is not intrinsic to a generator alone; it depends on the interaction between synthesis method, data structure, and downstream analytical task.

**Keywords:** Synthetic Data, Clustering, Regression, K-Means, Hierarchical Clustering, CART, Data Utility, Privacy.

# Resum


Les dades sintètiques s'utilitzen cada vegada més com a substitut de les dades originals per protegir la privadesa, però el seu valor depèn de si conserven les conclusions de les anàlisis posteriors. Aquesta tesi avalua aquesta utilitat analítica en dos contextos complementaris: el clustering, on la utilitat depèn de la fidelitat estructural i geomètrica, i la regressió, on la utilitat depèn de preservar coeficients, incertesa i intervals de confiança.

El primer estudi analitza si les dades sintètiques generades amb arbres de classificació i regressió (CART) conserven la geometria necessària per a tasques de clustering. Es comparen K-Means i clustering jeràrquic aglomeratiu en simulacions controlades que varien la mida mostral, el nombre de clústers, la separació, la dimensionalitat, la correlació i la família de distribucions. Els resultats mostren que CART és útil en escenaris normals i esfèrics, però introdueix artefactes en blocs que degraden el rendiment del clustering en dades Gamma esbiaixades. En conjunt, els dos mètodes de clustering mostren un comportament globalment similar sota aquestes distorsions sintètiques.

El segon estudi avalua la utilitat inferencial en regressió lineal i logística. Es compara la síntesi basada en CART amb generadors paramètrics mitjançant el solapament d'intervals de confiança, el ràtio de biaix i el ràtio d'amplada dels intervals. La síntesi paramètrica tendeix a obtenir els millors resultats quan les seves hipòtesis coincideixen amb el procés generador de dades, mentre que CART és menys fiable en escenaris d'alta dimensionalitat, alta correlació, mostres petites i regressió logística. En conjunt, la tesi mostra que la utilitat de les dades sintètiques no depèn només del generador, sinó de la interacció entre el mètode de síntesi, l'estructura de les dades i la tasca analítica posterior.



**Paraules clau:** Dades Sintètiques, Clustering, Regressió, K-Means, Clustering Jeràrquic, CART, Utilitat de les Dades, Privadesa.

# Resumen

Los datos sintéticos se utilizan cada vez más como sustituto de los datos originales para preservar la privacidad, pero su valor depende de si conservan las conclusiones de los análisis posteriores. Esta tesis evalúa esa utilidad analítica en dos contextos complementarios: el clustering, donde la utilidad depende de la fidelidad estructural y geométrica, y la regresión, donde la utilidad depende de preservar coeficientes, incertidumbre e intervalos de confianza.

El primer estudio analiza si los datos sintéticos generados mediante árboles de clasificación y regresión (CART) conservan la geometría necesaria para tareas de clustering. Se comparan K-Means y clustering jerárquico aglomerativo en simulaciones controladas que varían el tamaño muestral, el número de clústeres, la separación, la dimensionalidad, la correlación y la familia de distribuciones. Los resultados muestran que CART es útil en escenarios normales y esféricos, pero introduce artefactos en bloques que degradan el rendimiento del clustering en datos Gamma sesgados. En conjunto, ambos métodos de clustering muestran un comportamiento globalmente similar bajo estas distorsiones sintéticas.

El segundo estudio evalúa la utilidad inferencial en regresión lineal y logística. Se compara la síntesis basada en CART con generadores paramétricos usando el solapamiento de intervalos de confianza, el ratio de sesgo y el ratio de anchura de los intervalos. La síntesis paramétrica tiende a obtener mejores resultados cuando sus supuestos coinciden con el proceso generador de datos, mientras que CART es menos fiable en escenarios de alta dimensionalidad, alta correlación, muestras pequeñas y regresión logística. En conjunto, la tesis muestra que la utilidad de los datos sintéticos no depende solo del generador, sino de la interacción entre el método de síntesis, la estructura de los datos y la tarea analítica posterior.

**Palabras clave:** Datos Sintéticos, Clustering, Regresión, K-Means, Clustering Jerárquico, CART, Utilidad de los Datos, Privacidad.

# Acknowledgements

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Our collaboration evolved into a remarkably dynamic and ambitious partnership. What began as a focused investigation into clustering utility soon grew into a shared desire to push the boundaries of this research further. This mutual drive led us to expand the scope significantly, venturing into linear and then logistic regression to build a truly comprehensive evaluation. We consistently sought to exceed the initial expectations, resulting in a body of work that far transcends the standard 6-ECTS requirement. It has been a genuine privilege to work alongside someone who not only maintains a relentless pace but often sets it, constantly challenging me to reach higher levels of rigor and dedication. In an environment where finding such a matched tempo is rare, I am extremely proud of what we have achieved together through this constant, mutual push for excellence.

I would also like to thank Xinnuo Chen, whose preliminary exploratory work provided a solid foundation and inspired the systematic evaluation conducted in this study.

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# General Introduction

## 1.1 Motivation and Context

Data-driven research increasingly depends on access to detailed datasets, yet the same level of detail that makes data useful can also make it sensitive. In areas such as healthcare, finance, public administration, and social science, individual-level records often cannot be shared freely because of legal, ethical, and institutional constraints. Synthetic data has therefore become an important candidate for privacy-preserving analysis: instead of releasing the original records, a generator produces artificial observations intended to preserve the statistical properties needed for research [1]–[3].

The practical value of synthetic data, however, cannot be judged only by whether the generated records look plausible. The central question is whether an analyst using the synthetic data would reach conclusions similar to those obtained from the original data. This is the problem of analytical utility. Prior work distinguishes between broad similarity measures and task-specific utility measures, emphasizing that the right evaluation depends on the downstream analysis being supported [4].

This thesis studies synthetic-data utility from that downstream perspective. It asks whether common synthesis methods preserve enough information for two different families of machine-learning and statistical analysis:

- **clustering**, where the relevant utility is structural and geometric;
- **regression**, where the relevant utility is inferential and concerns coefficients, uncertainty, and confidence intervals.

## 1.2 Machine-Learning Perspective

Machine learning includes a broad set of methods that learn from data rather than relying on explicitly programmed rules. A common taxonomy distinguishes between supervised learning, unsupervised learning, and reinforcement learning. Figure 1.1 summarizes this classification and positions the two analytical settings studied in this thesis.

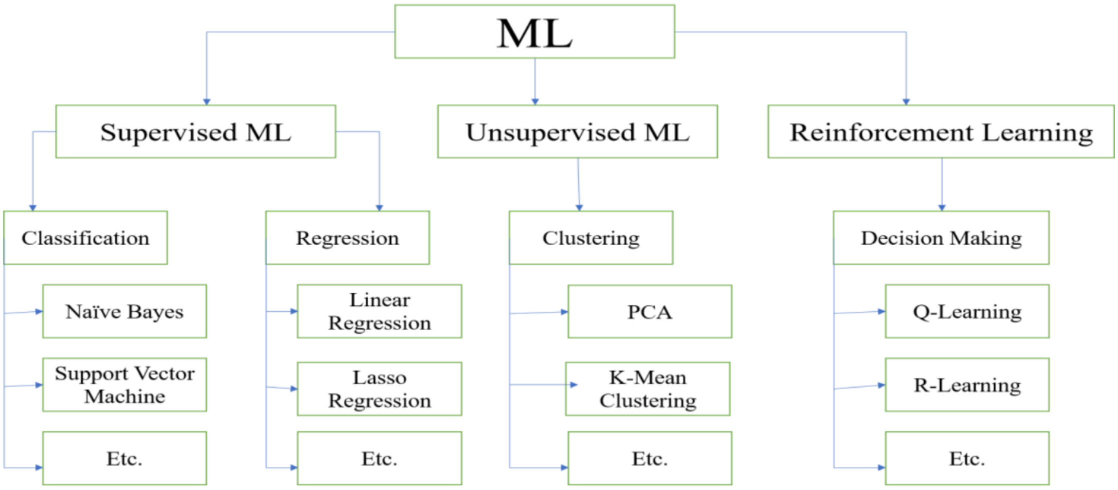


Figure 1.1: Taxonomy of common machine-learning paradigms and representative algorithms.

Source: Compiled by the author

Regression belongs to the supervised or model-based inferential setting, because it studies how an outcome changes as a function of predictors. Clustering belongs to unsupervised learning, because it attempts to discover structure without observing ground-truth labels during training. These two settings place different demands on synthetic data, as summarized in Table 1.1.

Table 1.1: Comparison of Structural and Inferential Utility demands on synthetic data.

| **Dimension** | **Structural Utility (Clustering)** | **Inferential Utility (Regression)** |
|---|---|---|
| **Learning Paradigm** | Unsupervised | Supervised / Model-based |
| **Primary Goal** | Preserve geometry and separability | Preserve relationships and uncertainty |
| **Key Features** | Distances, densities, topology | Point estimates, standard errors, CIs |
| **Sensitivity** | High sensitivity to spatial artifacts | High sensitivity to correlation limits |

Source: Compiled by the author based on [5],[6]

Clustering requires preservation of distances, densities, separability, and cluster geometry. Regression requires preservation of conditional relationships, point estimates, standard errors, and confidence intervals [5],[6].

## 1.3 Formal Problem Definition

The motivating assumption behind synthetic data is that a generated dataset can act as a useful substitute for the original data. However, the practical value of this substitution depends heavily on the "Privacy vs. Utility" trade-off. To generalize the problem and evaluate this trade-off rigorously, we must treat both clustering and regression as algorithms that extract a specific set of parameters or structural truths from a given dataset. By formalizing this extraction process, we can quantify the exact degradation introduced by synthesis.

### 1.3.1 The Core Unified Framework

We define the fundamental spaces and operators involved in the analytical lifecycle:

- **The Original Data:** Let $\mathcal{D}_R$ be the real (reference) dataset of size $N$ drawn from a true underlying joint probability distribution $P(X, Y)$ [3]. For unsupervised clustering tasks, $Y = \emptyset$, and the analysis focuses solely on the feature space $X$.

- **The Synthetic Generator:** Let $\mathcal{G}$ be a synthetic generation mechanism, such as the CART method which models variables via sequential conditional distributions. We define the resulting synthetic dataset as:

$$\mathcal{D}_S = \mathcal{G}(\mathcal{D}_R) \tag{1.1}$$

- **The Learning Algorithm:** Let $\mathcal{A}$ represent a machine learning algorithm applied to a dataset to extract a vector of learned properties or parameters, $\theta$.
  - *In Clustering:* $\mathcal{A}$ represents algorithms like K-Means or Hierarchical Clustering. The extracted parameters $\theta$ represent geometric structures, such as the number of clusters $k$, centroids $\mu$, and within-cluster variance $\sigma^2$.
  - *In Regression:* $\mathcal{A}$ represents supervised models like Linear or Logistic regression. Here, $\theta$ represents inferential relationships, primarily the estimated coefficients $\hat{\beta}$ and their associated standard errors or confidence intervals.

### 1.3.2 The Generalized Utility Discrepancy

The fundamental problem addressed in this thesis is evaluating the **Utility Discrepancy** caused by the generation mechanism $\mathcal{G}$. We aim to quantify the degradation between the true parameters learned from the real data, $\theta_R = \mathcal{A}(\mathcal{D}_R)$, and the parameters learned from the synthetic proxy, $\theta_S = \mathcal{A}(\mathcal{D}_S)$.

We define a task-specific discrepancy function $\Delta(\theta_R, \theta_S)$ that measures the loss of structural or inferential fidelity. The core research problem is to evaluate how $\Delta$ is influenced by the data space complexity, which is bounded by:

$$\Delta(\mathcal{A}(\mathcal{D}_R), \mathcal{A}(\mathcal{G}(\mathcal{D}_R))) = f(N, p, \rho, S) \tag{1.2}$$

Where the degradation is parameterized as a function of:

- $N$: The sample size of the dataset.
- $p$: The dimensionality (number of variables or predictors).
- $\rho$: The correlation structure or degree of cluster separation.
- $S$: The distributional shape or skewness (e.g., Normal vs. Gamma distributions).

### 1.3.3 Task-Specific Instantiations of the Discrepancy Function ($\Delta$)

To apply this generalized framework to the specific studies in the thesis, we instantiate $\Delta$ using the metrics tailored for each domain. These instantiations serve as the mathematical foundation for evaluating both unsupervised and supervised downstream tasks.

**A. The Clustering Instantiation (Geometric Fidelity)**

For unsupervised learning, the discrepancy function $\Delta$ measures topological and spatial distortion. The target parameters $\theta$ are the centroid locations ($\mu$) and the internal cluster variance. This thesis formalizes the discrepancy via the Mean Centroid Distance (MCD) and Mean Variance Difference (MVD). The MCD instantiation is defined as:

$$\Delta_{\text{MCD}}(\theta_R, \theta_S) = \frac{1}{k}\sum_{j=1}^{k} ||\mu_j^{(R)} - \mu_j^{(S)}||_2 \tag{1.3}$$

The analytical goal is to prove that for non-parametric generators like CART applied to skewed data ($S = \text{Gamma}$), $\Delta_{\text{MCD}}$ diverges significantly when the algorithm $\mathcal{A}$ requires convex/spherical assumptions (such as K-Means) compared to connectivity-based methods (such as Hierarchical Clustering).

**B. The Regression Instantiation (Inferential Fidelity)**

For supervised learning, $\Delta$ measures the distortion of conditional probabilities and statistical uncertainty. The target parameters $\theta$ are the point estimates ($\hat{\beta}$) and their Confidence Intervals ($[L, U]$). The discrepancy is formalized using the Bias Ratio ($B_R$) and Confidence Interval Overlap (CIO). To frame CIO as a penalty or discrepancy that should be minimized, we utilize $1 - \text{CIO}$:

$$\Delta_{\text{Inference}}(\theta_R, \theta_S) = 1 - \frac{1}{2}\left(\frac{\min(U_R, U_S) - \max(L_R, L_S)}{U_R - L_R} + \frac{\min(U_R, U_S) - \max(L_R, L_S)}{U_S - L_S}\right) \tag{1.4}$$

The analytical goal here is to demonstrate that sequentially compounding errors in the generator $\mathcal{G}$ over high dimensions ($p$) and high multicollinearity ($\rho$) artificially scales the variance of $\theta_S$, causing the inferential discrepancy to increase, particularly in small-sample logistic settings.

## 1.4 Objectives and Research Questions

The general objective of this thesis is to evaluate whether synthetic data preserves downstream analytical utility across two complementary settings: clustering and regression.

The specific objectives are:

1. To evaluate whether CART-generated synthetic data preserves the geometric structure required by clustering algorithms.

2. To compare the robustness of centroid-based and connectivity-based clustering under synthetic distortions.

3. To evaluate whether synthetic data preserves regression coefficients and confidence intervals in linear and logistic models.

4. To compare CART-based synthesis with parametric synthesis when the downstream analysis has a known model structure.

5. To identify conditions under which synthetic data remains useful, becomes unstable, or leads to misleading analytical conclusions.

These objectives are addressed through three thesis-level research questions:

- **RQ1:** Does synthetic data preserve the structural geometry required for unsupervised clustering?

- **RQ2:** Does synthetic data preserve the inferential quantities required for regression analysis?

- **RQ3:** How does the match between generator assumptions, data structure, and downstream analytical method determine synthetic-data utility?

## 1.5 Structure of the Thesis

The thesis is divided into two empirical studies under this common framing.

**Part I** studies structural utility for clustering. It evaluates K-Means and Agglomerative Hierarchical Clustering on CART-generated synthetic datasets, using controlled simulations and structural metrics such as cluster-number recovery, Gini coefficient, Mean Centroid Distance, and Mean Variance Difference.

**Part II** studies inferential utility for regression. It evaluates linear and logistic regression under CART and parametric synthesis, using Confidence Interval Overlap, Bias Ratio, and Confidence Interval Width Ratio to measure whether synthetic data supports the same statistical conclusions as the original data.

The final discussion integrates both studies and presents the overall thesis conclusion: synthetic data should be selected and validated according to the analysis it is expected to support, not treated as a universally interchangeable replacement for original data.

# Part II

# Clustering Study

# Chapter 2

# Clustering Utility Study

## 2.1 Purpose of the Study

Part I evaluates synthetic-data utility in the unsupervised-learning setting. Within the general thesis framework, clustering is used as a test of **structural utility**: whether a synthetic dataset preserves the geometry, separability, density, and dispersion required for an algorithm to recover meaningful groups.

The study focuses on synthetic data generated through the Classification and Regression Trees (CART) method implemented in the `synthpop` package. CART is flexible, but its axis-aligned tree partitions can turn smooth multivariate structure into block-like regions. This potential artifact is directly relevant to clustering because clustering algorithms depend on distances, shapes, and local connectivity.

## 2.2 Clustering as a Structural Utility Test

Unlike supervised methods, clustering algorithms do not have access to ground-truth labels during training. Their objective is to discover inherent structures by grouping similar observations together based on geometric or statistical properties.

Clustering is fundamental for exploratory data analysis, customer segmentation, and anomaly detection. However, because these algorithms rely heavily on the geometric structure of the data (distances, densities, and distributions), they are highly sensitive to data quality and distribution. Common algorithms include:

- **K-Means:** Partitioning data into $k$ groups by minimizing variance within clusters.
- **Hierarchical Clustering:** Building a tree of clusters (dendrogram) based on distance connectivity.

This part of the thesis extends the work of Xinnuo Chen [7] by systematically comparing how well K-Means and Agglomerative Hierarchical Clustering perform on CART-generated synthetic data.

We aim to answer three specific research questions:

1. **Detection Capability:** Can clustering algorithms correctly identify the optimal number of clusters ($k$) in synthetic data as well as they do in real data?

2. **Algorithm Robustness:** Which algorithm is more resilient to the distortions introduced by the synthetic generation process? We hypothesise that Hierarchical Clustering may be more robust than K-Means in non-normal distributions.

3. **Quantification of Degradation:** To what extent does the CART generation method degrade the geometric properties (separability, density) of the clusters?

To answer these questions, we have developed a reproducible pipeline that generates 1,800+ datasets varying in sample size ($N$), cluster separation, noise levels, and probability distributions (Normal vs. Gamma), allowing for a rigorous statistical comparison.

## 2.3 Structure of Part I

This part is organized as follows:

- **Chapter 3: Methods** details the mathematical formulations of K-Means and Hierarchical Clustering and describes the synthetic data generation via CART. It also defines the structural fidelity metrics and the simulation pipeline used to evaluate algorithm robustness.

- **Chapter 4: Results** presents the empirical analysis, including PCA and t-SNE visualizations alongside cluster detection success rates. It further quantifies geometric degradation using the Gini coefficient, Mean Centroid Distance, and Mean Variance Difference metrics.

- **Chapter 5: Conclusions** synthesizes the clustering findings, highlighting the shared sensitivity of both clustering methods to synthetic geometric distortion and identifying the practical implications for structural utility evaluation.

# Chapter 3

# Clustering Methods

The methodology adopted in this research is divided into three main stages: the mathematical definition of the clustering algorithms selected for comparison, the generation of synthetic data using the CART method, and the simulation pipeline designed to evaluate the structural degradation of the data.

## 3.1 Clustering Algorithms

Clustering is the task of grouping a set of objects in such a way that objects in the same group (cluster) are more similar to each other than to those in other groups. This research focuses on two of the most established paradigms in unsupervised learning: centroid-based clustering (K-Means) and connectivity-based clustering (Hierarchical).

### 3.1.1 K-Means Clustering

K-Means is a partition-based algorithm that aims to divide $N$ observations into $k$ non-overlapping clusters. It operates by minimizing the within-cluster sum of squares (WCSS), also known as inertia [8].

**Formal Mathematical Definition**

We define the K-Means procedure by the tuple

$$\mathcal{M}_{KM} = \langle \mathcal{X}, k, \mathcal{P}, \boldsymbol{\mu}, \mathcal{A}, \mathcal{U}, J \rangle \tag{3.1}$$

where each component is defined as follows:

- **Data Space ($\mathcal{X}$):** The observed sample is

$$\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}, \qquad \mathbf{x}_i \in \mathbb{R}^d \tag{3.2}$$

  with $N$ observations and feature dimension $d$.

- **Number of Clusters ($k$):** A fixed integer satisfying

$$k \in \mathbb{N}, \qquad 1 < k < N \tag{3.3}$$

that specifies the target cardinality of the partition.

- **Partition Space ($\mathcal{P}$):** A feasible clustering is a partition

$$\mathcal{P} = \{S_1, S_2, \ldots, S_k\} \tag{3.4}$$

such that

$$S_a \cap S_b = \varnothing \quad (a \neq b), \qquad \bigcup_{j=1}^{k} S_j = \mathcal{X}, \qquad S_j \neq \varnothing \tag{3.5}$$

- **Centroid Set ($\boldsymbol{\mu}$):** Each cluster $S_j$ is represented by a centroid

$$\boldsymbol{\mu} = \{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_k\}, \qquad \boldsymbol{\mu}_j \in \mathbb{R}^d \tag{3.6}$$

with

$$\boldsymbol{\mu}_j = \frac{1}{|S_j|} \sum_{\mathbf{x}_i \in S_j} \mathbf{x}_i \tag{3.7}$$

- **Assignment Operator ($\mathcal{A}$):** For fixed centroids $\boldsymbol{\mu}^{(t)}$ at iteration $t$, the assignment map is

$$\mathcal{A}_{\boldsymbol{\mu}^{(t)}} : \mathcal{X} \to \{1, \ldots, k\}, \qquad \mathcal{A}_{\boldsymbol{\mu}^{(t)}}(\mathbf{x}_i) = \arg \min_{1 \leq j \leq k} \|\mathbf{x}_i - \boldsymbol{\mu}_j^{(t)}\|_2^2 \tag{3.8}$$

and therefore induces the partition

$$S_j^{(t)} = \left\{\mathbf{x}_i \in \mathcal{X} : \mathcal{A}_{\boldsymbol{\mu}^{(t)}}(\mathbf{x}_i) = j\right\} \tag{3.9}$$

- **Update Operator ($\mathcal{U}$):** For a fixed partition $\mathcal{P}^{(t)} = \{S_1^{(t)}, \ldots, S_k^{(t)}\}$, the centroid update is

$$\mathcal{U}_{\mathcal{P}^{(t)}}(S_j^{(t)}) = \boldsymbol{\mu}_j^{(t+1)} = \frac{1}{|S_j^{(t)}|} \sum_{\mathbf{x}_i \in S_j^{(t)}} \mathbf{x}_i, \qquad j = 1, \ldots, k \tag{3.10}$$

- **Objective Function ($J$):** The algorithm minimizes the within-cluster sum of squares

$$J(\mathcal{P}, \boldsymbol{\mu}) = \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in S_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|_2^2 \tag{3.11}$$

which is equivalently the inertia of the partition.

The iterative state of the algorithm can be written as

$$\left(\mathcal{P}^{(t)}, \boldsymbol{\mu}^{(t)}\right) \mapsto \left(\mathcal{P}^{(t+1)}, \boldsymbol{\mu}^{(t+1)}\right) \tag{3.12}$$

with the alternating minimization steps

$$\mathcal{P}^{(t+1)} = \arg\min_{\mathcal{P}} J(\mathcal{P}, \boldsymbol{\mu}^{(t)}), \qquad \boldsymbol{\mu}^{(t+1)} = \arg\min_{\boldsymbol{\mu}} J(\mathcal{P}^{(t+1)}, \boldsymbol{\mu}) \tag{3.13}$$

The convergence criterion can be written as either

$$\mathcal{P}^{(t+1)} = \mathcal{P}^{(t)} \qquad \text{or} \qquad \|\boldsymbol{\mu}^{(t+1)} - \boldsymbol{\mu}^{(t)}\|_F \leq \varepsilon \tag{3.14}$$

for some tolerance $\varepsilon > 0$, where $\|\cdot\|_F$ denotes the Frobenius norm.

**Time and Space Complexity**

Let $I$ denote the number of iterations until convergence. The main computational blocks are:

$$T_{\text{init}} = O(kd) \tag{3.15}$$

for selecting or storing the initial centroids,

$$T^{(t)}_{\text{assign}} = O(Nkd) \tag{3.16}$$

for computing the distances from all $N$ points to all $k$ centroids at iteration $t$, and

$$T^{(t)}_{\text{update}} = O(Nd) \tag{3.17}$$

for recomputing all centroids once the assignments are known.

Hence the total running time is

$$T_{KM}(N, k, d, I) = T_{\text{init}} + \sum_{t=1}^{I} \left(T^{(t)}_{\text{assign}} + T^{(t)}_{\text{update}}\right) = O(kd) + O(I(Nkd + Nd)) \tag{3.18}$$

which is dominated by

$$T_{KM}(N, k, d, I) = O(NkId) \tag{3.19}$$

The corresponding memory requirement is

$$S_{KM}(N, k, d) = O(Nd) + O(kd) + O(N) \tag{3.20}$$

for the data matrix, the centroid matrix, and the assignment vector, which simplifies to

$$S_{KM}(N, k, d) = O(Nd + kd) \tag{3.21}$$

**Practical Considerations**

K-Means is highly efficient in practice, but it has significant limitations relevant to this study:

- It assumes clusters are spherical and of similar size.

- It is sensitive to outliers and the initial placement of centroids.
- The number of clusters $k$ must be specified *a priori*.

### 3.1.2 Hierarchical Clustering (Agglomerative)

Unlike K-Means, Hierarchical Clustering does not require a pre-specified number of clusters. Instead, it builds a hierarchy of clusters, often represented as a dendrogram. This research utilizes **Agglomerative Clustering** (bottom-up), which treats each observation as its own cluster and successively merges pairs of clusters [9].

**Formal Mathematical Definition (Ward's Linkage)**

Agglomerative Hierarchical Clustering with Ward's linkage is a deterministic, greedy, bottom-up agglomerative framework. It can be formally defined by the tuple

$$\mathcal{M}_{HC} = \langle \mathcal{X}, \mathcal{P}^{(0)}, \mathcal{T}, \Delta, \mathcal{L}, \mathcal{M} \rangle \tag{3.22}$$

where:

- **Data Space ($\mathcal{X}$):** An input set of $N$ observations

$$\mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \subset \mathbb{R}^d \tag{3.23}$$

- **Initial Partition ($\mathcal{P}^{(0)}$):** The algorithm starts from the singleton partition

$$\mathcal{P}^{(0)} = \{\{\mathbf{x}_1\}, \{\mathbf{x}_2\}, \ldots, \{\mathbf{x}_N\}\} \tag{3.24}$$

- **Hierarchy ($\mathcal{T}$):** A dendrogram represented as a sequence of nested partitions

$$\mathcal{T} = \{\mathcal{P}^{(0)}, \mathcal{P}^{(1)}, \ldots, \mathcal{P}^{(N-1)}\} \tag{3.25}$$

  satisfying

$$|\mathcal{P}^{(t)}| = N - t, \qquad t = 0, 1, \ldots, N-1 \tag{3.26}$$

  and each $\mathcal{P}^{(t+1)}$ is obtained by merging exactly two blocks from $\mathcal{P}^{(t)}$.

- **Error Sum of Squares ($\Delta$):** The spatial dispersion within any candidate cluster $C \subset \mathcal{X}$ is quantified by the Error Sum of Squares (ESS), defining the function $\Delta(C)$:

$$\Delta(C) = \mathrm{ESS}(C) = \sum_{\mathbf{x} \in C} ||\mathbf{x} - \boldsymbol{\mu}_C||_2^2 \quad \text{where} \quad \boldsymbol{\mu}_C = \frac{1}{|C|} \sum_{\mathbf{x} \in C} \mathbf{x} \tag{3.27}$$

- **Linkage Objective Function ($\mathcal{L}$):** The merging cost between two disjoint clusters $A$ and $B$ is defined by Ward's minimal variance criterion. The algorithm greedily selects the pair $(A^*, B^*)$ that minimizes the incremental increase in the total intra-cluster variance:

$$\mathcal{L}(A,B) = \Delta(A \cup B) - [\Delta(A) + \Delta(B)] = \frac{|A||B|}{|A|+|B|}||\boldsymbol{\mu}_A - \boldsymbol{\mu}_B||_2^2 \tag{3.28}$$

At each agglomerative step $t$, the transition $\mathcal{P}^{(t-1)} \to \mathcal{P}^{(t)}$ operates via:

$$(A_t^*, B_t^*) = \arg \min_{A,B \in \mathcal{P}^{(t-1)}, A \neq B} \mathcal{L}(A,B) \tag{3.29}$$

and the merged partition becomes

$$\mathcal{P}^{(t)} = \left(\mathcal{P}^{(t-1)} \setminus \{A_t^*, B_t^*\}\right) \cup \{A_t^* \cup B_t^*\} \tag{3.30}$$

- **Merge Operator ($\mathcal{M}$):** The agglomeration rule may be written as

$$\mathcal{M}(\mathcal{P}^{(t-1)}) = \mathcal{P}^{(t)} \tag{3.31}$$

so that the full algorithm is the recursive application of $\mathcal{M}$ for $t = 1, \ldots, N-1$.

**Equivalent Lance–Williams Update for Ward's Linkage**

If $A$ and $B$ are merged into $A \cup B$, then for any other active cluster $C$, the updated Ward distance can be written in Lance–Williams form as

$$\begin{aligned} d_W(A \cup B, C) = \frac{|A|+|C|}{|A|+|B|+|C|} d_W(A,C) + \frac{|B|+|C|}{|A|+|B|+|C|} d_W(B,C) \\ - \frac{|C|}{|A|+|B|+|C|} d_W(A,B) \end{aligned} \tag{3.32}$$

which shows explicitly how the pairwise merge costs are updated after each agglomerative step.

**Time and Space Complexity**

Let $m_t = N - t + 1$ denote the number of active clusters at step $t$. In a naive implementation, the dominant tasks are:

$$T_{\text{dist}} = O(N^2 d) \tag{3.33}$$

to construct the initial pairwise dissimilarity structure,

$$T_{\text{search}}^{(t)} = O(m_t^2) \tag{3.34}$$

to search for the minimum merge cost among all active cluster pairs, and

$$T^{(t)}_{\text{update}} = O(m_t) \tag{3.35}$$

to update the linkage values after a merge.

Therefore, the total cost can be expressed as

$$T_{HC}(N, d) = O(N^2 d) + \sum_{t=1}^{N-1} \Big( O(m_t^2) + O(m_t) \Big) \tag{3.36}$$

and since $\sum_{t=1}^{N-1} m_t^2 = O(N^3)$, this yields

$$T_{HC}(N, d) = O(N^2 d + N^3) = O(N^3) \tag{3.37}$$

in the standard naive formulation.

With optimized data structures, the practical running time can be reduced to approximately

$$T^{\text{opt}}_{HC}(N, d) = O(N^2 \log N) \quad \text{or} \quad O(N^2) \tag{3.38}$$

depending on the specific implementation.

The storage requirement is dominated by the pairwise dissimilarity representation:

$$S_{HC}(N, d) = O(Nd) + O(N^2) \tag{3.39}$$

which simplifies asymptotically to

$$S_{HC}(N, d) = O(N^2) \tag{3.40}$$

**Practical Considerations**

Hierarchical clustering is computationally expensive, making it less suitable for massive datasets. However, it offers distinct advantages:

- It captures hierarchical structures and does not assume strictly spherical clusters.
- It is deterministic (produces the same result every time), unlike K-Means which depends on random initialization.
- It allows for the determination of $k$ *post-hoc* by cutting the dendrogram at a specific height.

### 3.1.3 Comparative and Complexity Summary

When selecting a clustering algorithm for synthetic data evaluation, understanding the computational constraints is as critical as understanding the geometric assumptions.

**Time Complexity:** The asymptotic contrast between the two methods can be summarized as

$$T_{KM}(N, k, d, I) = O(NkId) \qquad \text{versus} \qquad T_{HC}(N, d) = O(N^3) \tag{3.41}$$

for the standard forms of the algorithms. Since $k$, $I$, and $d$ are usually much smaller than $N$ in our simulation settings, K-Means is often treated as effectively linear in $N$, whereas Hierarchical Clustering remains superlinear because it must maintain and update inter-cluster dissimilarities throughout the agglomerative process.

**Computational (Space) Complexity:** The corresponding memory requirements are

$$S_{KM}(N, k, d) = O(Nd + kd) \qquad \text{and} \qquad S_{HC}(N, d) = O(N^2) \tag{3.42}$$

so K-Means scales linearly with the sample representation, while Hierarchical Clustering is dominated by the quadratic cost of storing and updating pairwise distances. This creates the main computational tradeoff between the two methods.

Table 3.1 summarizes the key computational, structural, and operational differences between the two algorithms evaluated in this study.

Table 3.1: Comparison of K-Means and Hierarchical Clustering

| **Feature** | **K-Means** | **Hierarchical (Ward)** |
|---|---|---|
| **Time Complexity** | $O(N \cdot k \cdot I \cdot d)$ (Linear) | $O(N^2 \log N)$ to $O(N^3)$ |
| **Space Complexity** | $O(N \cdot d + k \cdot d)$ (Linear) | $O(N^2)$ (Quadratic) |
| **Geometry** | Assumes spherical, convex clusters | Flexible, based on connectivity |
| **Parameters** | Requires $k$ beforehand | $k$ determined by cutting tree |
| **Robustness** | Sensitive to noise/outliers | More robust (structure-based) |
| **Deterministic** | No (Random initialization) | Yes |

Source: Compiled by the author

## 3.2 Data Generation Framework

To rigorously evaluate algorithm performance, we implemented a reverse-engineering pipeline that generates a "Ground Truth" (Real) dataset and subsequently creates a Synthetic version using the method under investigation.

### 3.2.1 Reference Data Generation (Real)

The reference datasets were generated using the `mvtnorm` package in R [10],[11] to simulate controlled multivariate clusters. We varied four key parameters to create a total of 180 unique data configurations. Each configuration was then repeated $M = 100$ times; in the experiment JSON reported in Appendix A.3, this replicate count is stored under the field `m`:

- **Sample Size ($N$):** Ranging from 100 to 1,000 observations to test scalability.
- **Number of Clusters ($k$):** Set to 2, 3, or 4 to simulate varying levels of complexity.
- **Cluster Separation:** A separation index controlling the overlap between clusters. Higher separation implies distinct, easy-to-detect clusters; lower separation implies significant overlap.
- **Probability Distribution:**
  - **Normal:** Standard Gaussian clusters (spherical).
  - **Gamma:** Skewed distributions (non-spherical) to test the robustness of the algorithms against non-normal geometries [12].

### 3.2.2 Synthetic Data Generation (CART Method)

For each real dataset, a corresponding synthetic dataset was generated using the **Classification and Regression Trees (CART)** method, implemented in the `synthpop` R package [3].

For the qualitative visualization figures presented later in Chapter 4, one additional clarification is necessary. The leftmost synthetic panels reuse simulation-side cluster colouring only as a diagnostic device to compare the generated point cloud with the known reference geometry. These colours do **not** imply a one-to-one correspondence between original and synthetic records, and such "true labels" would not be observable in a realistic synthetic-data release. Accordingly, those panels are interpreted only as simulated diagnostics; the substantive evaluation relies on the clustering predictions and quantitative metrics reported below.

**The Sequential Regression Approach**

The CART method synthesizes data variable by variable. Let $X = (X_1, X_2, ..., X_p)$ be the vector of variables in the real dataset. The joint distribution is approximated as a product of conditional distributions:

$$f(X_1, ..., X_p) \approx f(X_1) \cdot f(X_2|X_1) \cdot f(X_3|X_1, X_2) \cdot ... \cdot f(X_p|X_1, ..., X_{p-1}) \tag{3.43}$$

The synthesis proceeds sequentially:

1. **Step 1:** Generate $X_1$ by resampling from the original univariate distribution.
2. **Step 2:** Train a Decision Tree (CART) to predict $X_2$ using $X_1$ as the predictor.

3. **Step 3:** Generate synthetic values for $X_2$ by passing the synthetic $X_1$ through the tree and sampling from the appropriate leaf node.

4. **Step 4:** Repeat for all subsequent variables, using all previously generated variables as predictors.

**Structural Artifacts**

While CART is non-parametric and can capture non-linear relationships, it introduces specific geometric artifacts. Because decision trees split data using orthogonal hyperplanes (rules like $x > 5$), they tend to approximate smooth diagonal curves with "staircase" or "blocky" boundaries. A key objective of this study is to determine if this "blocky" structure causes K-Means or Hierarchical Clustering to fail.

## 3.3 Evaluation Metrics

Standard clustering accuracy metrics (like purity) are insufficient when the cluster labels are unknown or permuted. We introduced a set of structural fidelity metrics to quantify the degradation between Real and Synthetic data. Table 3.2 provides a summary of the metrics employed.

Table 3.2: Summary of the geometric and detection metrics used to evaluate clustering utility.

| **Metric** | **Description** | **Target** |
|---|---|---|
| **Success Rate** | Binary measure of whether the algorithm detected the true $k$. | 1.0 (100%) |
| **Gini Coefficient** | Measures the inequality of cluster sizes (imbalance). | $\approx 0.0$ |
| **MCD** | Mean Centroid Displacement: shift in cluster centers. | $\approx 0.0$ |
| **MVD** | Mean Variance Difference: change in cluster dispersion. | $\approx 0.0$ |

Source: Compiled by the author

### 3.3.1 Preprocessing Step: Cluster Alignment (Hungarian Algorithm)

It is crucial to note that cluster alignment is not an evaluation metric *per se*, but rather a mandatory preprocessing step required before any comparative structural fidelity metrics can be applied. Because clustering algorithms are fundamentally unsupervised, the cluster labels they assign are arbitrary (e.g., “Cluster 1” in the Original Data might be labeled “Cluster 3” in the Synthetic Data).

To calculate accuracy or geometric distortion between the two datasets, we must first find the optimal 1-to-1 mapping between the original labels and the synthetic labels. We achieve this by formulating it as the **Linear Sum Assignment Problem** and solving it using the **Hungarian Algorithm** [13].

We construct a cost matrix $C$ where $C_{ij}$ represents the number of mismatches if Predicted Cluster $i$ is assigned to True Cluster $j$. The algorithm finds the optimal permutation of labels that minimizes this cost (maximizes the overlap), allowing the subsequent metrics (like MCD and MVD) to be calculated on correctly matched pairs.

### 3.3.2 Success Rate (Detection)

The primary metric is binary: **Did the algorithm find the correct number of clusters?** We utilize the *Silhouette Score and the Elbow Method* (for K-Means) [14] or the *Dendrogram Cut* (for Hierarchical) to estimate $\hat{k}$.

$$\text{Success} = \begin{cases} 1 & \text{if } \hat{k}_{\text{predicted}} = k_{\text{true}} \\ 0 & \text{otherwise} \end{cases} \tag{3.44}$$

### 3.3.3 Structural Fidelity Metrics

To measure the geometric distortion introduced by synthesis, we calculate the following comparative metrics between the Real and Synthetic clusters:

**1. Gini Coefficient (Cluster Balance)**

The Gini coefficient measures the inequality of cluster sizes. If the synthetic data generation collapses a small cluster into a large one, the Gini coefficient will increase.

$$G = \frac{\sum_{i=1}^{k}\sum_{j=1}^{k}|n_i - n_j|}{2k\sum_{i=1}^{k} n_i} \tag{3.45}$$

Where $n_i$ is the number of points in cluster $i$. A value of 0 indicates perfectly equal cluster sizes; a value near 1 indicates high imbalance. To evaluate the structural distortion introduced by the synthesis process, we compute the absolute difference between the Gini coefficient of the Synthetic Data ($G^{(S)}$) and the Original Data ($G^{(R)}$):

$$\Delta G = |G^{(S)} - G^{(R)}| \tag{3.46}$$

A $\Delta G$ near 0 indicates that the synthetic data has successfully preserved the original cluster balance.

**2. Mean Centroid Displacement (MCD)**

We measure how far the centers of the clusters have shifted during synthesis. Let $\mu_j^{(R)}$ be the centroid of cluster $j$ in Real data and $\mu_j^{(S)}$ in Synthetic data:

$$\text{MCD} = \frac{1}{k}\sum_{j=1}^{k} ||\mu_j^{(R)} - \mu_j^{(S)}||_2 \tag{3.47}$$

High displacement indicates that the synthetic data has drifted from the original spatial location.

**3. Mean Variance Difference (MVD)**

We measure if the synthetic clusters are more dispersed (noisy) or more compact than the real ones. We compare the average within-cluster variance ($\sigma^2$):

$$\text{MVD} = \frac{1}{k}\sum_{j=1}^{k} \left|\text{Var}(C_j^{(R)}) - \text{Var}(C_j^{(S)})\right| \tag{3.48}$$

# Chapter 4

# Clustering Results

## 4.1 Visualization and Dimensionality Reduction

To visually assess the geometric distortions introduced by the synthetic generation process, we require methods to project high-dimensional data into a lower-dimensional space. This dimensionality reduction eases the direct 2D visual comparison between the Original Data (OD) and the Synthetic Data (SD), allowing us to intuitively identify structural artifacts before proceeding to quantitative metrics.

### 4.1.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) projects the data linearly, preserving the global geometric structure and highlighting the orthogonal "blocky" artifacts introduced by the decision-tree-based synthesis [15]. Figure 4.1 shows the PCA projection for the Normal scenario, and Figure 4.2 shows the corresponding Gamma scenario. In both figures, the leftmost synthetic panels reuse simulation-side colouring only as a diagnostic comparison aid; they do not represent observable true labels for synthetic records in practice.

**Normal Distribution**

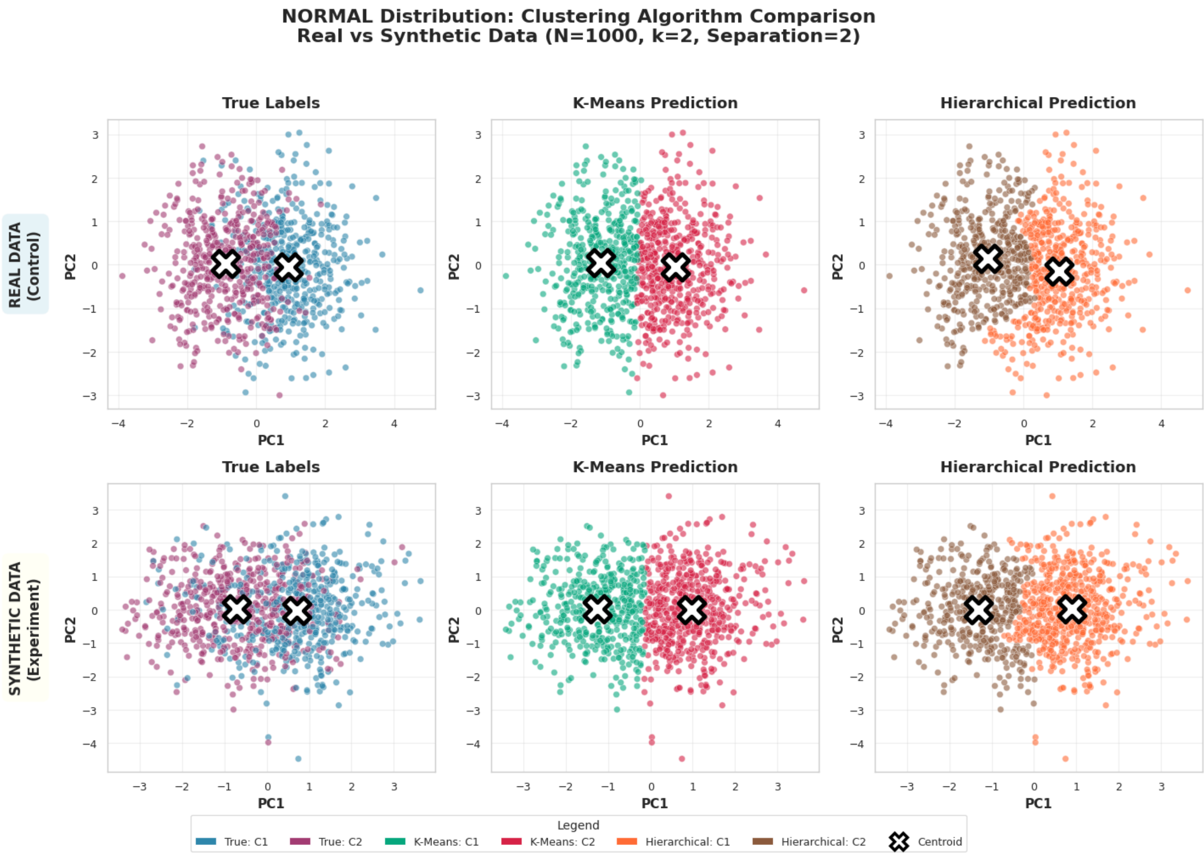


Figure 4.1: **PCA Projection: Normal Distribution (**$N = 1000, k = 2$**).** Comparison of Real (Top) vs. Synthetic (Bottom). In the lower-left panel, "True Labels" denotes simulation-side diagnostic colouring only and does not imply observable labels or recordwise correspondence in practice.

Source: Compiled by the author

In the Normal distribution case, Figure 4.1 shows that the spherical cluster geometry is largely preserved after synthesis. *Observation:* The diagnostic colouring of the synthetic sample remains broadly aligned with the original structure, and the K-Means and Hierarchical Clustering panels recover nearly the same left-right partition, indicating high utility for normally distributed synthetic data.

### Gamma Distribution

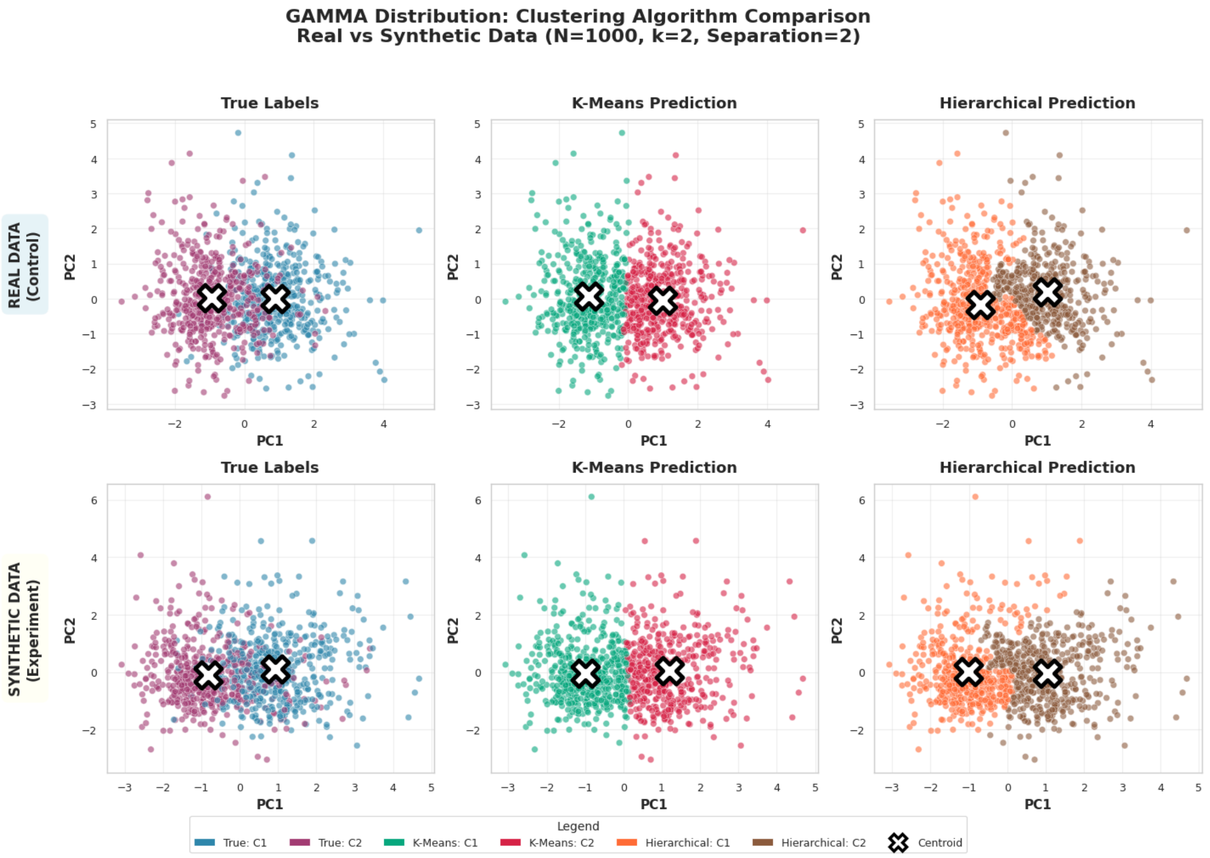


Figure 4.2: **PCA Projection: Gamma Distribution ($N = 1000, k = 2$).** Comparison of Real (Top) vs. Synthetic (Bottom). In the lower-left panel, "True Labels" denotes simulation-side diagnostic colouring only and does not imply observable labels or recordwise correspondence in practice.

Source: Compiled by the author

In the Gamma distribution case, Figure 4.2 reveals the geometric degradation more clearly. *Observation:* The diagnostic colouring of the synthetic sample suggests "hardened" orthogonal boundaries compared with the smooth tails of the real data. K-Means fails to adapt to this distorted geometry, imposing a linear decision boundary that cuts through the cluster tails, whereas Hierarchical Clustering proves more robust.

### 4.1.2 t-SNE Analysis

t-SNE provides a non-linear projection that emphasizes the separation between manifolds [16]. This visualization is particularly effective at highlighting the "bad cuts" made by centroid-based algorithms on skewed data. Figure 4.3 presents the t-SNE embedding for the Normal scenario, while Figure 4.4 presents the Gamma scenario. As with the PCA figures, the leftmost synthetic panels are diagnostic colourings retained from the simulation workflow and are not available to an analyst in a practical synthetic-data release.

**Normal Distribution**

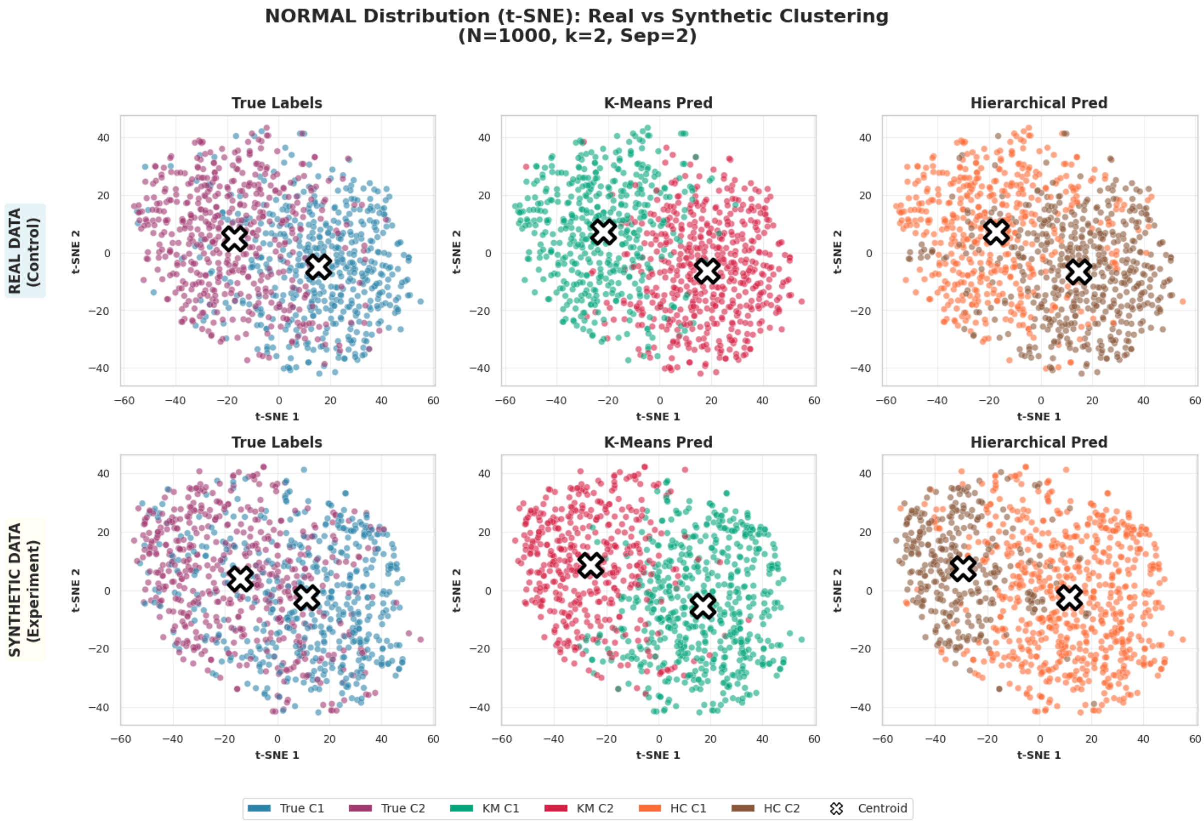


Figure 4.3: **t-SNE Projection: Normal Distribution.** Comparison of Real (Top) vs. Synthetic (Bottom). In the lower-left panel, "True Labels" denotes simulation-side diagnostic colouring only and does not imply observable labels or recordwise correspondence in practice.

Source: Compiled by the author

Figure 4.3 shows that, even in the non-linear projection, the Normal distribution remains cohesive and separable after synthesis. *Observation:* The synthetic-data embedding maintains two distinct groups similar to the control sample. Both K-Means and Hierarchical Clustering produce nearly identical partitions, confirming that the synthesis process does not introduce major topological distortions in spherical distributions.

**Gamma Distribution** The Gamma distribution was included to explore how a certain degree of asymmetry in the data affects clustering results on synthetic datasets.

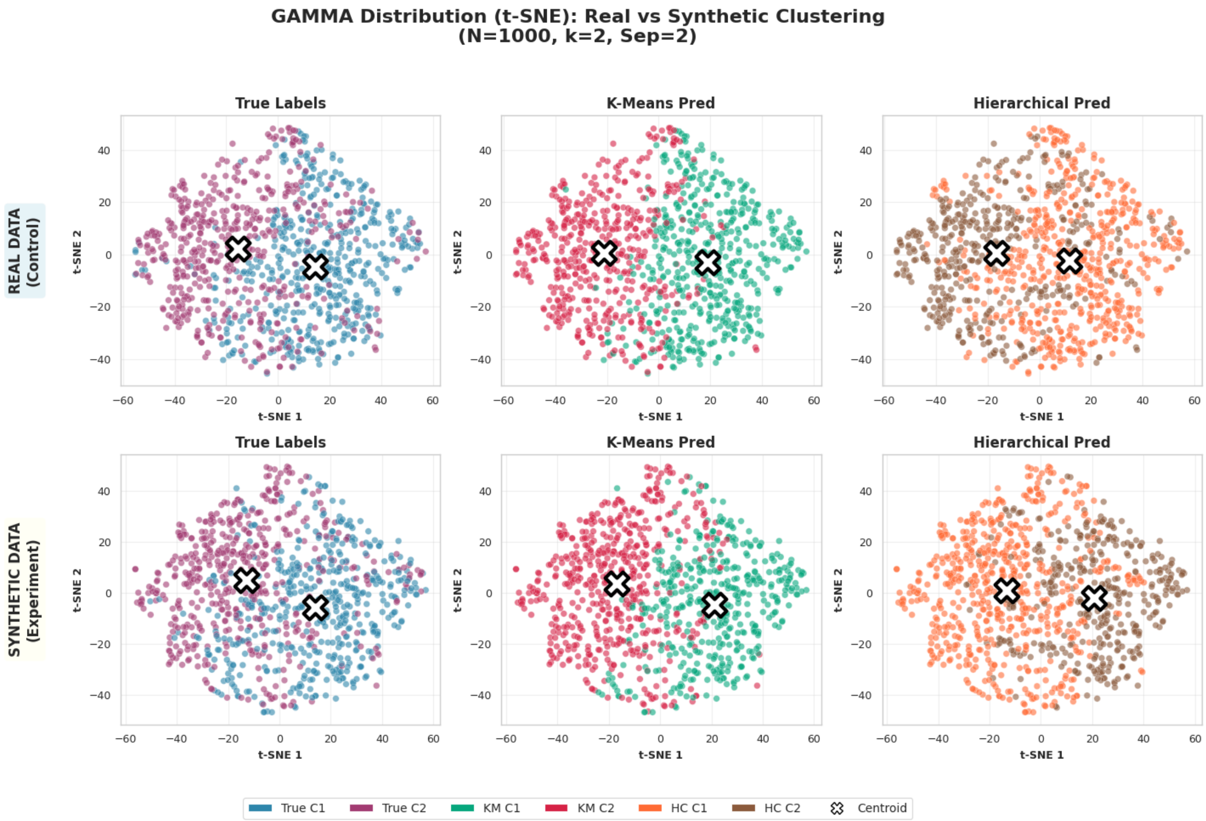


Figure 4.4: **t-SNE Projection: Gamma Distribution.** Comparison of Real (Top) vs. Synthetic (Bottom). In the lower-left panel, "True Labels" denotes simulation-side diagnostic colouring only and does not imply observable labels or recordwise correspondence in practice.

Source: Compiled by the author

Figure 4.4 reveals the main failure mode of K-Means on skewed synthetic data. *Critical Finding:* In the synthetic row, the diagnostic colouring suggests two connected density masses. K-Means arbitrarily slices these masses with a rigid boundary, failing to capture their organic shape. In contrast, Hierarchical Clustering follows the connected structure more closely, offering a substantially better qualitative approximation.

## 4.2 Cluster Number Matching

This section analyzes the "Success Rate" metric—defined as the proportion of synthetic datasets where the algorithm correctly estimated the true number of clusters ($k$). We compare the performance of K-Means and Hierarchical Clustering across varying cluster separations and cluster counts.

For a given configuration with $M$ total simulations, the **Success Rate (SR)** is calculated as the proportion of correct detections.

**Normal Distribution** Figure 4.5 summarizes the success-rate surface for the Normal scenarios under both clustering algorithms.

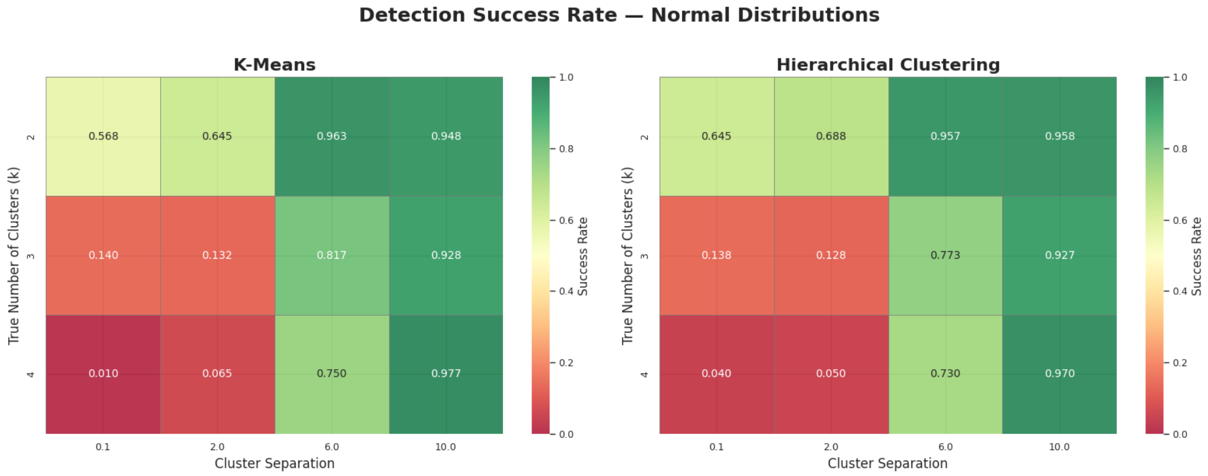


Figure 4.5: **Success Rate on Normal Distribution.** Left: K-Means. Right: Hierarchical Clustering. Green indicates high success; Red indicates failure.

Source: Compiled by the author

**Observations:**

- **High Separation Stability:** For well-separated clusters, both algorithms achieve near-perfect performance. The synthetic generation artifacts do not hinder detection when the clusters are naturally distinct. Note that the approximate radius of each cluster is close to 2. Therefore, a distance equal to 6 indicates that there is a blank space of about 2 units between clusters, ensuring no overlap.

- **Low Separation Degradation:** A sharp performance drop is observed when separation is very low. This suggests that for spherical, overlapping clusters, the "blocky" noise from synthetic generation affects both algorithms equally.

- **Effect of the Number of Clusters:** Figure 4.5 also suggests slightly better recovery when the true number of clusters is small, as the detection surfaces are more stable for simpler two-cluster settings than for scenarios with more clusters.

- **Algorithm Parity:** In this distribution, the color patterns between K-Means (Left) and Hierarchical Clustering (Right) are largely symmetrical, indicating no significant advantage for either method on spherical data.

**Gamma Distribution** Figure 4.6 reports the analogous success-rate surface for the Gamma scenarios.

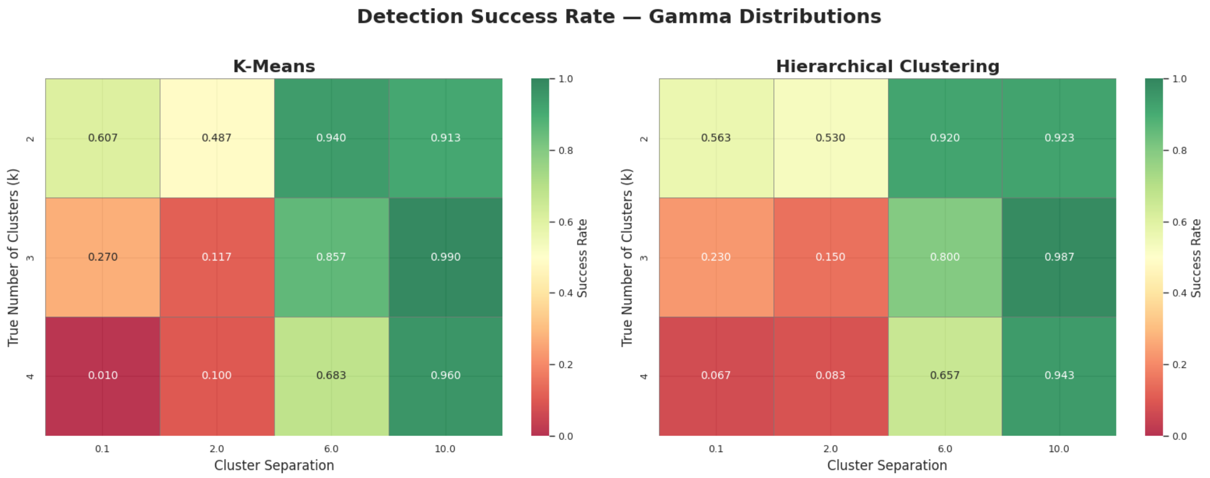


Figure 4.6: **Success Rate on Gamma Distribution.** Left: K-Means. Right: Hierarchical Clustering. Note the prevalence of red/orange zones in the K-Means heatmap.

Source: Compiled by the author

**Observations:**

- **K-Means Collapse:** The K-Means heatmap exhibits a dominant "Red/Orange" zone extending into moderate separation levels. Detection is not guaranteed even at higher separations.

- **Comparable Behavior:** The Hierarchical Clustering heatmap shows a pattern broadly similar to K-Means, with both methods degrading as the design becomes less favorable.

- **Performance Delta:** Comparing the two matrices visually suggests only limited differences between the algorithms. For skewed data, the main pattern is that both methods become less reliable as the synthetic geometry departs from the original cluster structure.

## 4.3 Gini Coefficient

To assess structural fidelity, we analyzed the Gini coefficient, which measures the inequality of cluster sizes. We compared this index between the real and synthetic data.

**Normal Distribution** Figure 4.7 displays the absolute Gini-coefficient distribution for the Normal scenarios.

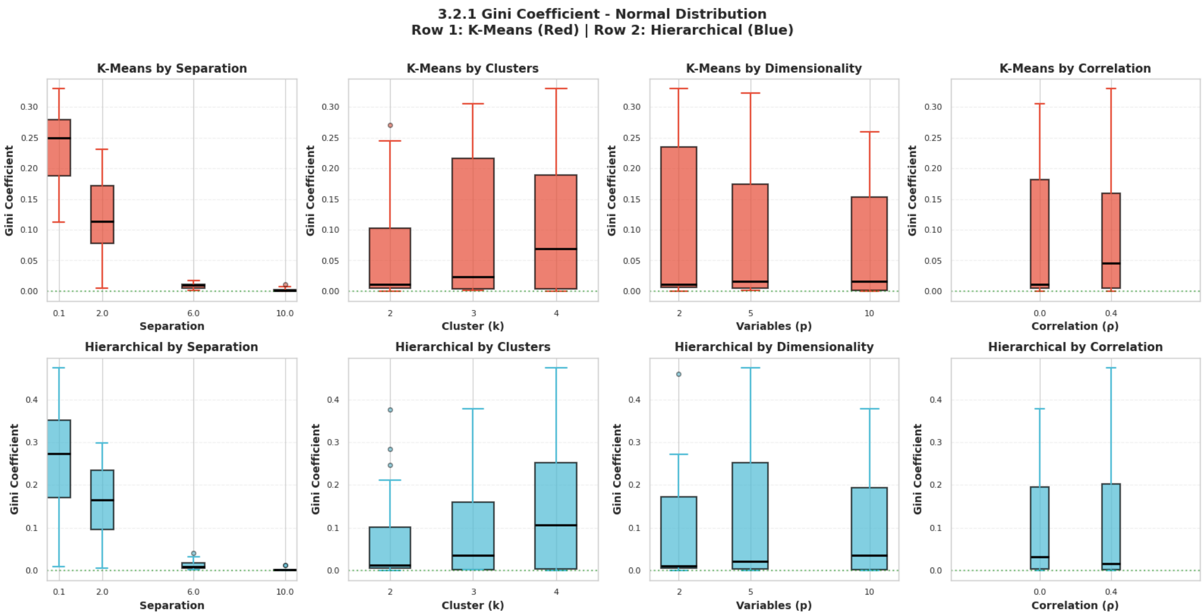


Figure 4.7: **Absolute difference between Gini coefficients (Synthetic vs. Real).** Normal Distribution. Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric approaches zero as separation increases, indicating balanced detection.

Source: Compiled by the author

**Sensitivity to Separation:** Both algorithms exhibit a strong dependency on cluster distinctness. At low separation, the synthetic noise dominates, leading to arbitrary partitions and high imbalance. As separation increases, both algorithms converge to near-perfect balance, confirming that the synthetic generation process does not inherently disrupt cluster balance for well-separated spherical data.

**Dimensionality Stability:** While K-Means maintains a tight distribution near zero for high dimensions, Hierarchical Clustering exhibits a wider interquartile range and a higher median Gini, suggesting less stability in maintaining balance as feature space grows.

**Gamma Distribution** Figure 4.8 displays the corresponding absolute Gini-coefficient distribution for the Gamma scenarios.

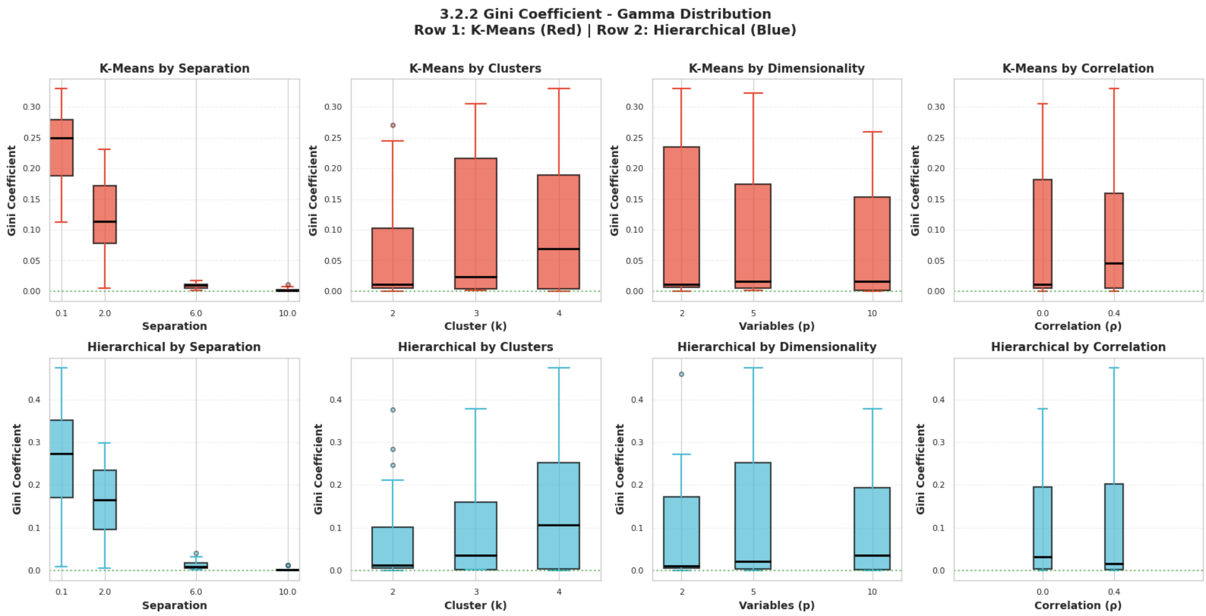


Figure 4.8: **Absolute difference between Gini coefficients (Synthetic vs. Real).** Gamma Distribution. Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). Note the high variance in Hierarchical Clustering compared to the rigid stability of K-Means.

Source: Compiled by the author

**K-Means Rigidity:** A striking observation is the consistently low Gini coefficients produced by K-Means. Despite the data being skewed and non-spherical, K-Means minimizes within-cluster variance by imposing a Voronoi tessellation that slices the data into roughly equal volumes. This effectively "forces" the skewed data into a balanced structure, suppressing the natural geometric properties of the Gamma distribution.

**Hierarchical Flexibility (and Instability):** In contrast, Hierarchical Clustering exhibits significantly higher Gini coefficients and wider interquartile ranges. This indicates that Hierarchical Clustering is less constrained by the assumption of equal cluster sizes, allowing it to form imbalanced groups that may better reflect the dense "heads" and diffuse "tails" of the Gamma distribution, though at the cost of higher variance.

**Comparative Analysis:** While K-Means remains structurally insensitive to the changing geometry (maintaining a flat, low-Gini profile), Hierarchical Clustering reacts dynamically. This confirms that for non-normal synthetic data, **K-Means introduces a "balancing bias"**, while Hierarchical Clustering retains the underlying (imbalanced) structural signal.

## 4.4 Mean Centroid Distance

We computed the Mean Centroid Distance (MCD) to quantify the spatial shift between the cluster centers in the real dataset versus the synthetic dataset. Before computing this metric, the clusters in the synthetic and real datasets must first be paired using the Hungarian Algorithm described in Chapter 3, Section 3.3.

**Normal Distribution** Figure 4.9 reports the Mean Centroid Distance for the Normal scenarios.

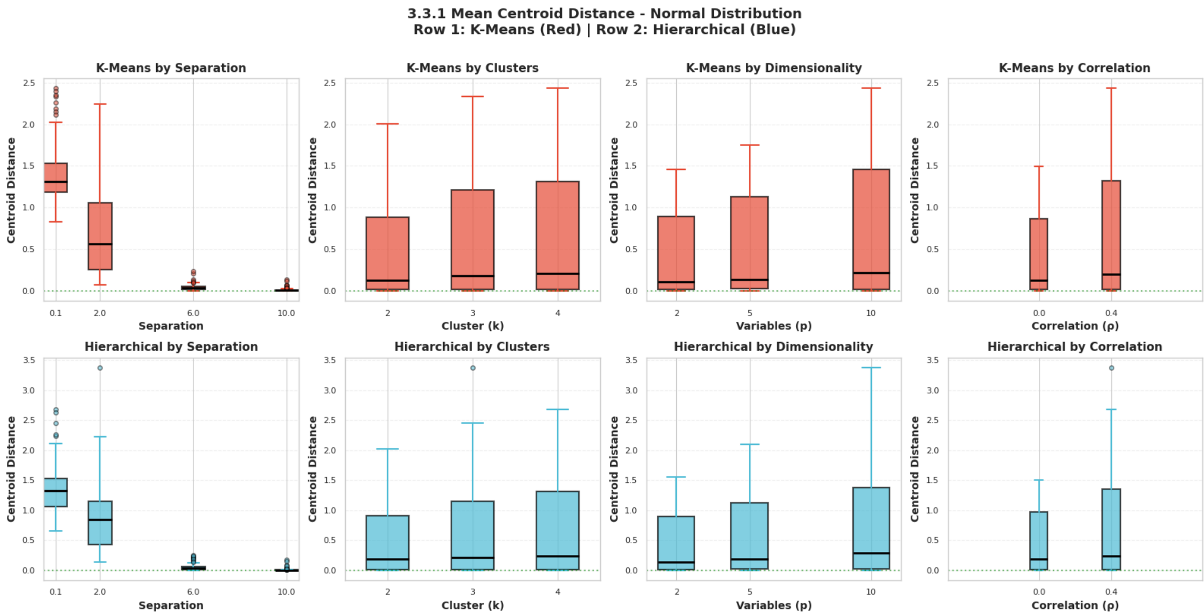


Figure 4.9: **Mean Centroid Distance (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric quantifies the Euclidean shift of cluster centers after synthesis.

Source: Compiled by the author

The results show a dominant inverse relationship between cluster separation and centroid stability. At the lowest separation level, both algorithms show high displacement, indicating significant centroid drift when clusters overlap. As separation increases, the distances for both algorithms effectively collapse to zero. In terms of data complexity, the variance of the centroid distance increases with both the number of clusters ($k$) and dimensionality ($p$).

**Gamma Distribution** Figure 4.10 reports the Mean Centroid Distance for the Gamma scenarios.

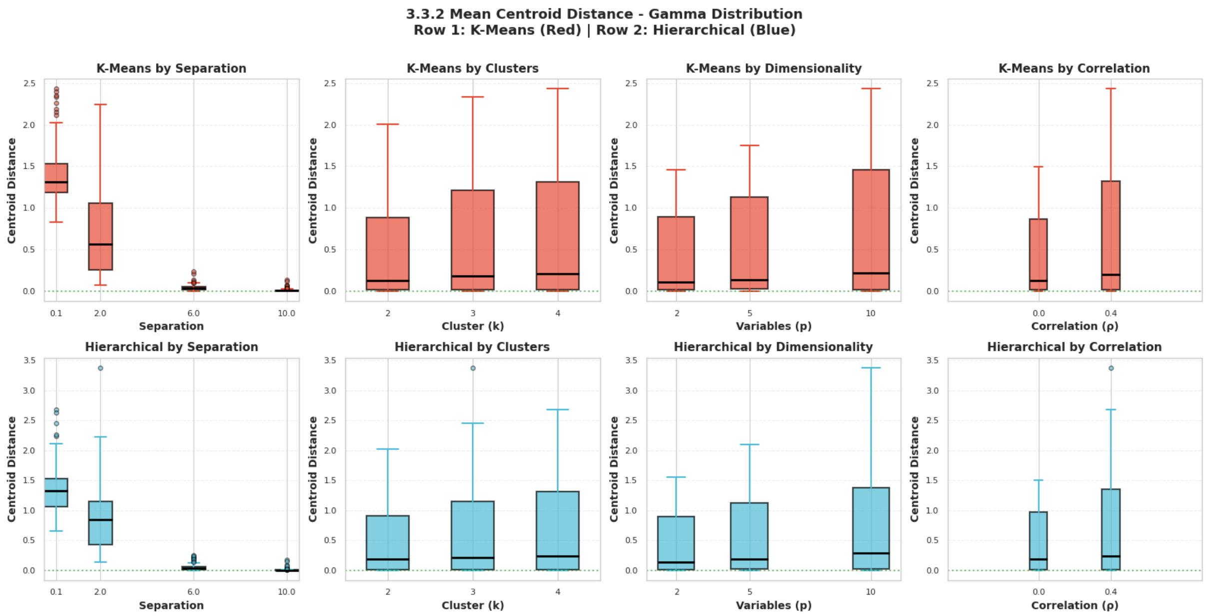


Figure 4.10: **Mean Centroid Distance (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue).

Source: Compiled by the author

The overall trends closely mirror those observed in the Normal distribution, suggesting that centroid drift is primarily driven by the "blocky" artifacts of the CART synthesis method rather than the specific probability distribution. However, the impact of complexity is more pronounced here. Higher dimensionality and a higher number of clusters coincide with increased centroid displacement.

## 4.5 Mean Variance Difference

We calculated the Mean Variance Difference (MVD) to measure the discrepancy in within-cluster dispersion between the real and synthetic datasets.

**Normal Distribution** Figure 4.11 presents the Mean Variance Difference for the Normal scenarios.

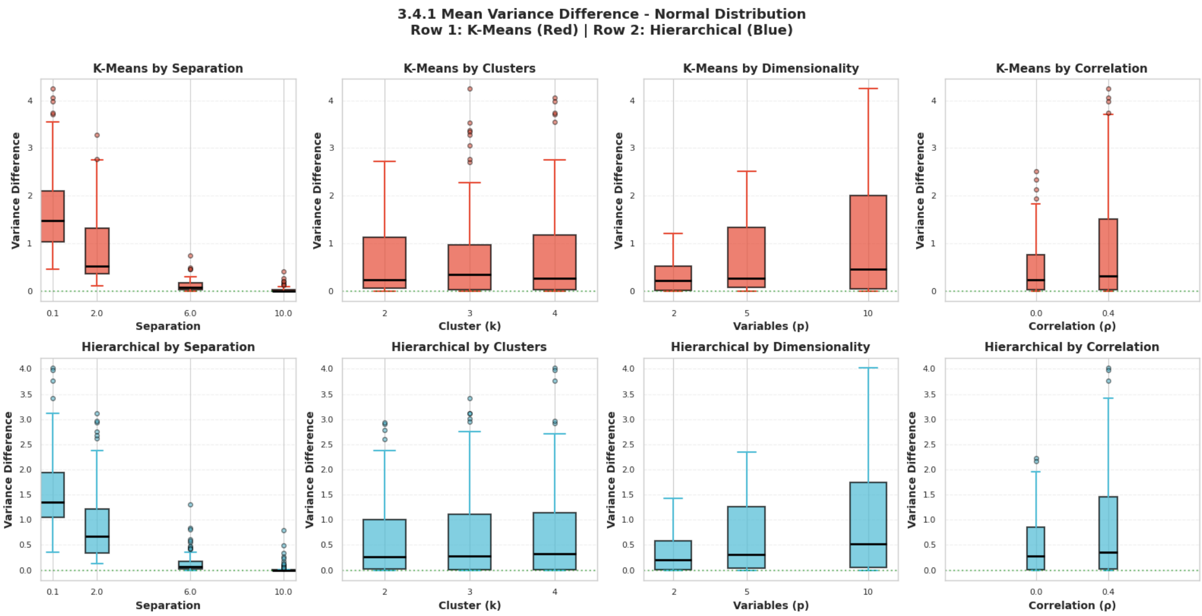


Figure 4.11: **Mean Variance Difference (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric increases with dimensionality and feature correlation.

Source: Compiled by the author

The results indicate that cluster separation is the primary factor reducing variance distortion; at high separation, the difference approaches zero for both algorithms. However, structural complexity negatively impacts fidelity. Increasing the number of variables or introducing correlation significantly increases the variance difference. This suggests that the synthetic generation process struggles to replicate the exact dispersion of multivariate normal clusters in higher-dimensional or correlated spaces.

**Gamma Distribution** Figure 4.12 presents the corresponding Mean Variance Difference for the Gamma scenarios.

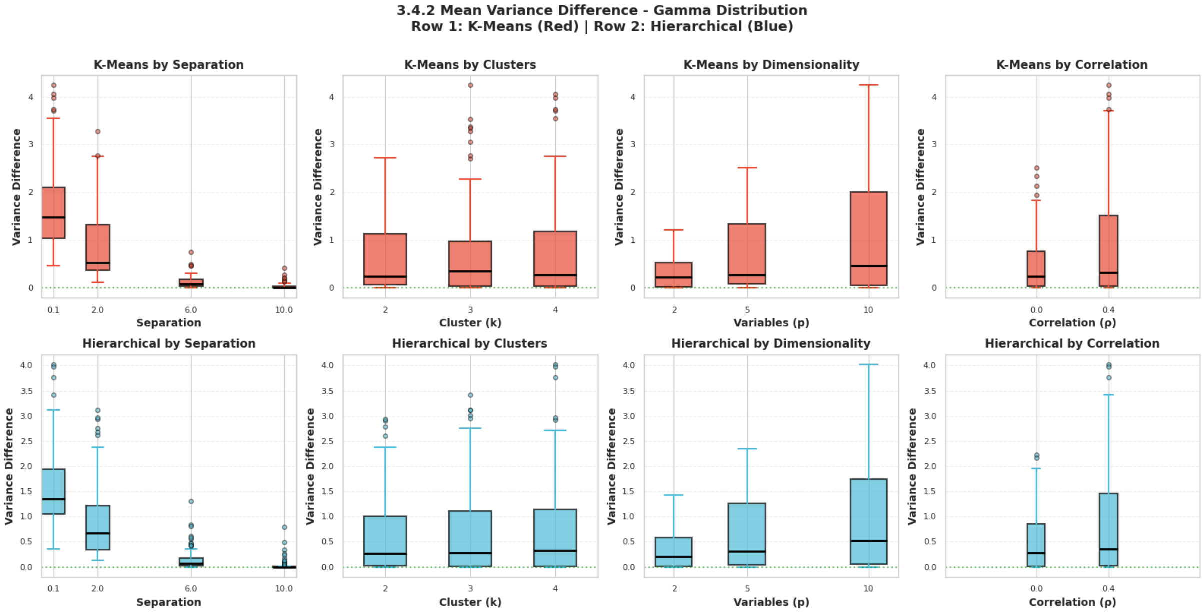


Figure 4.12: **Mean Variance Difference (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). Similar to the Normal case, variance distortion peaks in high-dimensional and correlated settings.

Source: Compiled by the author

The variance discrepancy patterns closely track those observed in the Normal distribution, indicating that dispersion artifacts are driven more by the synthetic generation method than the underlying probability distribution. Furthermore, the “by Dimensionality” and “by Feature Correlation” panels confirm that maintaining accurate cluster variance is increasingly difficult as complexity grows, leading to elevated difference metrics for both algorithms....

# Chapter 5

# Clustering Conclusions

This research set out to evaluate the "Privacy vs. Utility" trade-off in the context of unsupervised learning, specifically questioning whether synthetic data generated via the Classification and Regression Trees (CART) method retains sufficient geometric fidelity for accurate clustering. By analyzing over 1,800 datasets across varying distributions, noise levels, and dimensionalities, we have quantified the structural degradation introduced by synthesis and its impact on K-Means and Hierarchical Clustering.

## 5.1 Key Findings

The empirical results show that the CART synthesis method introduces specific "blocky" orthogonal artifacts due to the nature of decision tree splits. While these artifacts are limited in spherical Normal distributions, they become more visible in skewed Gamma distributions, where the smooth tails of clusters are replaced by more angular synthetic boundaries.

Within this setting, K-Means and Hierarchical Clustering display broadly similar behavior when recovering cluster structure from the synthetic data. Both algorithms perform well when the geometry is simple and the clusters are well separated, and both deteriorate when the synthetic geometry becomes more distorted or the design becomes more complex. Rather than identifying a clear winner, the results suggest that the dominant source of degradation is the synthesis process itself.

The structural fidelity metrics reinforce this interpretation. The Gini analysis indicates that K-Means tends to produce more balanced partitions in skewed settings, whereas Hierarchical Clustering allows more imbalance to remain visible. However, this difference does not translate into a consistent overall advantage in cluster identification. In both methods, centroid drift and variance distortion increase in high-dimensional, overlapping, or otherwise difficult scenarios, confirming that the main limitation lies in preserving the multivariate structure during synthesis rather than in one clustering algorithm clearly outperforming the other.

Across the simulated designs, performance improved as cluster separation increased, and it worsened as the number of clusters, the number of variables, or the correlation between variables increased. In other words, greater separation made the synthetic clustering task easier, whereas more complex cluster structure and feature dependence made it harder for the synthetic data to preserve the geometry needed by either clustering method.

## 5.2 Comparison with previous Work

This study extends the foundational work of Xinnuo Chen [7], which established the baseline utility of synthetic data. While previous studies largely focused on supervised learning tasks, such as classification accuracy, or global statistical properties, this research specifically addressed the unsupervised domain. Unlike general utility metrics that average performance over a dataset, our introduction of structural metrics—specifically the Mean Centroid Distance and Gini Coefficient—allowed for a granular diagnosis of why algorithms fail. We demonstrated that utility is not a binary property of the data, but an interaction between the generation method and the analysis method. More cautiously, the visual evidence in Figure 4.4 suggests that the orthogonal artifacts introduced by CART can interfere with the smooth geometric assumptions often associated with K-Means in non-normal settings, but this effect should be interpreted as a plausible mechanism rather than as a definitive incompatibility unique to K-Means.

## 5.3 Limitations

The scope of this research implies certain limitations that must be considered when interpreting the results. First, we exclusively evaluated the CART method implemented in the `synthpop` package. Other synthesis techniques, such as parametric methods or deep learning approaches, may produce different geometric artifacts that could interact differently with clustering algorithms. Second, the comparison was strictly limited to K-Means and Agglomerative Hierarchical Clustering. Density-based algorithms like DBSCAN or model-based clustering methods were not tested, though they may offer distinct advantages in handling synthetic noise. Finally, the use of pure Normal and Gamma distributions provided a controlled experimental setting but may not fully capture the chaotic noise, outliers, and mixed-type variables often found in real-world administrative data.

## 5.4 Generalization

Despite the specific algorithms used, the findings generalize to the broader point that clustering utility depends on how well the synthetic data preserves the geometry required by the downstream method. Algorithms based on different geometric assumptions may react differently to CART-induced distortions, but the present results do not support a strong ranking between centroid-based and connectivity-based approaches. This is crucial for data scientists selecting tools for privacy-preserving analytics, as it suggests that validation against the intended clustering workflow is just as critical as the data quality itself.

## 5.5 Future Research Directions

To further resolve the friction between data privacy and analytical utility, future research should focus on alternative generation methods. Evaluating whether Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs) can generate smooth, non-orthogonal decision boundaries would be a logical next step to see if they preserve the convexity required for K-Means. Additionally, there is significant potential in developing metric-optimized synthesis loss functions that explicitly penalize the distortion of centroid distances or cluster variance during the training of the generator. Finally, creating a standardized "Unsupervised Utility Score" based on the structural metrics defined in this study could help automatically flag synthetic datasets that are unsuitable for clustering tasks before they are released to researchers.

## 5.6 Conclusions

A synthesized comparison of the algorithmic robustness under CART generation is presented in Table 5.1.

Table 5.1: Summary of algorithmic robustness to CART-induced geometric artifacts.

| Condition | K-Means | Hierarchical (Ward) |
|---|---|---|
| **Normal Data (Spherical)** | Highly Robust | Highly Robust |
| **Gamma Data (Skewed)** | Similar degradation patterns | Similar degradation patterns |
| **High Dimensionality** | High Centroid Drift | High Centroid Drift |
| **Overlapping Clusters** | Poor Detection | Poor Detection |

Source: Compiled by the author

The promise of synthetic data is that it offers "privacy by design" without compromising analytical truth. This research clarifies the boundaries of that promise. We conclude that synthetic data generated via CART is not utility-neutral; it can alter the geometry of the data in ways that affect cluster recovery under both K-Means and Hierarchical Clustering, with no clear evidence here that one of the two methods is systematically superior.

For organizations leveraging synthetic data, the practical recommendation is therefore not to privilege one of these clustering algorithms in advance, but to validate the chosen method against the geometry preserved by the synthetic release. When the original distribution is simple and well separated, both K-Means and Hierarchical Clustering remain viable. When the data are skewed, overlapping, high-dimensional, or highly correlated, both methods should be interpreted more cautiously because the main risk arises from the synthetic distortion itself. By understanding these geometric interactions, researchers can more effectively unlock the value of sensitive data while maintaining rigorous privacy standards.

# Part III

# Regression Study

# Chapter 6

# Regression Utility Study

## 6.1 Regression as an Inferential Utility Test

Part II evaluates synthetic-data utility in the regression setting. Within the general thesis framework, regression is used as a test of **inferential utility**: whether synthetic data preserves the statistical conclusions that would be obtained from the original data.

Regression analysis remains the most widely used statistical tool for inferential research. Researchers rely on linear and logistic models to quantify relationships, test hypotheses, and inform policy. If synthetic data is to be used as a proxy for real data in academic and professional publications, it must preserve not only the marginal distributions but also the conditional relationships (coefficients) and their associated uncertainty (standard errors and confidence intervals) [5],[6].

Most synthetic data evaluations focus on utility through predictive accuracy or distributional similarity. However, for inference, the primary concern is whether the same scientific conclusion would be reached on synthetic data as on original data [4]. This requires specialized metrics such as Confidence Interval Overlap (CIO) and a systematic understanding of how data complexity, sample size, dimensionality, and correlation affect inferential fidelity.

## 6.2 Objectives and Research Questions

The primary objective of this part of the study is to evaluate the inferential utility of synthetic data generators across a wide range of simulated regression scenarios, determining under which conditions the different methods preserve the coefficients and confidence intervals of the original data most effectively.

The specific objectives are:

1. To compare the performance of CART-based synthesis against parametric generators (Normal, Logistic) in linear and logistic regression models.

2. To quantify the impact of sample size ($N$), number of predictors ($p$), and correlation structure ($\rho$) on the quality of synthetic inference.

3. To identify the specific conditions under which synthetic data leads to biased or overly optimistic/pessimistic conclusions.

The regression study is guided by the following research questions:

- **RQ1:** How reliably can synthetic data reproduce the confidence intervals of a linear regression model compared to a logistic one?

- **RQ2:** Does increasing the dimensionality ($p$) or complexity of the correlation structure ($\rho$) significantly degrade the CIO?

- **RQ3:** How does the relative performance of CART-based and parametric synthesis change across different regression settings?

## 6.3 Scope and Contributions

This study covers univariate and multivariate simulations for linear and logistic regression. It evaluates synthesis methods provided by the `synthpop` package, specifically focusing on the CART (Classification and Regression Trees), Normal, and Logistic regression synthesis arms.

The study focuses exclusively on **sequential synthetic data generation (1 to 1)**, meaning variables are synthesized one by one conditioned on previously generated ones. It does not cover joint/simultaneous generation methods, deep learning approaches (GANs, VAEs), or explicit privacy-risk measurements (e.g., membership inference attacks).

The study provides a systematic benchmarking of common synthesis methods across 1,800+ scenarios. It introduces a structured evaluation framework based on CIO, Bias Ratio, and Width Ratio, offering practical guidance for researchers on which generator to choose depending on their data's characteristics.

## 6.4 Structure of Part II

This part is organized as follows: Chapter 7 describes the experimental design and metrics; Chapter 8 presents the results for linear models; Chapter 9 details the logistic regression findings; and Chapter 10 synthesizes the regression conclusions.

# Chapter 7

# Regression Methods

## 7.1 Regression Framework

### 7.1.1 Linear Regression

The linear regression model is the foundation for analyzing the relationship between a continuous outcome $Y$ and a set of predictors $\mathbf{X} = \{X_1, X_2, \ldots, X_p\}$ [17]. We define the linear-regression model by the tuple

$$\mathcal{M}_{LR} = \langle \mathcal{X}, \mathbf{y}, \tilde{\mathbf{X}}, \boldsymbol{\beta}, \varepsilon, f, \ell, \hat{\boldsymbol{\beta}}, \widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) \rangle \tag{7.1}$$

where:

- $\mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ is the predictor sample, with $\mathbf{x}_i \in \mathbb{R}^p$.

- $\mathbf{y} = (y_1, \ldots, y_N)^\top \in \mathbb{R}^N$ is the response vector.

- $\tilde{\mathbf{X}} \in \mathbb{R}^{N \times (p+1)}$ is the design matrix including the intercept column,

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N1} & \cdots & x_{Np} \end{bmatrix} \tag{7.2}$$

- $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)^\top \in \mathbb{R}^{p+1}$ is the coefficient vector.

- $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_N)^\top$ is the error vector, with

$$\varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \tag{7.3}$$

- $f$ denotes the conditional mean map

$$f(\mathbf{x}_i; \boldsymbol{\beta}) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} \tag{7.4}$$

The model itself is

$$\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \varepsilon \tag{7.5}$$

or, component-wise,

$$Y_i = \beta_0 + \sum_{j=1}^{p} \beta_j X_{ij} + \varepsilon_i \tag{7.6}$$

Under the Gaussian assumption, the conditional density is

$$Y_i \mid \mathbf{x}_i \sim \mathcal{N}\Big(\mathbf{x}_i^* \boldsymbol{\beta}, \sigma^2\Big) \tag{7.7}$$

where $\mathbf{x}_i^* = (1, x_{i1}, \ldots, x_{ip})$ denotes the augmented predictor row.

The likelihood and log-likelihood are

$$L(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \tilde{\mathbf{X}}) = (2\pi\sigma^2)^{-N/2} \exp\Big(-\frac{1}{2\sigma^2}(\mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta})^\top(\mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta})\Big) \tag{7.8}$$

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta})^\top(\mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta}) \tag{7.9}$$

The ordinary least squares estimator is

$$\hat{\boldsymbol{\beta}}_{OLS} = (\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^\top\mathbf{y} \tag{7.10}$$

provided that $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$ is nonsingular.

Its estimated covariance matrix is

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}_{OLS}) = \hat{\sigma}^2(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1} \tag{7.11}$$

with residual variance estimate

$$\hat{\sigma}^2 = \frac{1}{N - p - 1}\|\mathbf{y} - \tilde{\mathbf{X}}\hat{\boldsymbol{\beta}}_{OLS}\|_2^2 \tag{7.12}$$

**Time and Space Complexity**

The main computational blocks are:

$$T_{\text{crossprod}} = O(Np^2), \qquad T_{\text{solve}} = O(p^3), \qquad T_{\text{fitted}} = O(Np) \tag{7.13}$$

corresponding to the construction of $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$, inversion/linear solve, and fitted-value computation. Therefore,

$$T_{LR}(N, p) = O(Np^2 + p^3) \tag{7.14}$$

and the memory requirement is

$$S_{LR}(N, p) = O(Np + p^2) \tag{7.15}$$

### 7.1.2 Logistic Regression

For binary outcomes $Y \in \{0, 1\}$, we employ the logistic regression model [18]. We define it by the tuple

$$\mathcal{M}_{LogR} = \langle \mathcal{X}, \mathbf{y}, \tilde{\mathbf{X}}, \boldsymbol{\beta}, g, \boldsymbol{\pi}, L, \ell, \hat{\boldsymbol{\beta}}, \mathcal{I}(\hat{\boldsymbol{\beta}}) \rangle \tag{7.16}$$

where:

- $\mathcal{X}$, $\mathbf{y}$, $\tilde{\mathbf{X}}$, and $\boldsymbol{\beta}$ are defined as above, except now $y_i \in \{0, 1\}$.
- $g(\cdot)$ is the logit link
$$g(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) \tag{7.17}$$
- $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)^\top$ is the success-probability vector, with
$$\pi_i = P(Y_i = 1 \mid \mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^* \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^* \boldsymbol{\beta})} \tag{7.18}$$

The systematic component is

$$\eta_i = \mathbf{x}_i^* \boldsymbol{\beta} = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} \tag{7.19}$$

and the link relation is

$$\operatorname{logit}(\pi_i) = \eta_i \tag{7.20}$$

The response model is

$$Y_i \mid \mathbf{x}_i \sim \operatorname{Bernoulli}(\pi_i) \tag{7.21}$$

with likelihood

$$L(\boldsymbol{\beta} \mid \mathbf{y}, \tilde{\mathbf{X}}) = \prod_{i=1}^{N} \pi_i^{y_i} (1 - \pi_i)^{1 - y_i} \tag{7.22}$$

and log-likelihood

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{N} [y_i \log(\pi_i) + (1 - y_i) \log(1 - \pi_i)] \tag{7.23}$$

The score vector is

$$U(\boldsymbol{\beta}) = \frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \tilde{\mathbf{X}}^\top (\mathbf{y} - \boldsymbol{\pi}) \tag{7.24}$$

and the observed Fisher information is

$$\mathcal{I}(\boldsymbol{\beta}) = \tilde{\mathbf{X}}^\top \mathbf{W} \tilde{\mathbf{X}}, \qquad \mathbf{W} = \operatorname{diag}(\pi_i(1-\pi_i))_{i=1}^N \tag{7.25}$$

The maximum likelihood estimator $\hat{\boldsymbol{\beta}}$ solves

$$U(\hat{\boldsymbol{\beta}}) = \mathbf{0} \tag{7.26}$$

and is typically computed via Newton–Raphson / IRLS updates:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \left(\tilde{\mathbf{X}}^\top \mathbf{W}^{(t)} \tilde{\mathbf{X}}\right)^{-1} \tilde{\mathbf{X}}^\top (\mathbf{y} - \boldsymbol{\pi}^{(t)}) \tag{7.27}$$

The asymptotic covariance estimator is

$$\widehat{\operatorname{Var}}(\hat{\boldsymbol{\beta}}) = \mathcal{I}(\hat{\boldsymbol{\beta}})^{-1} = \left(\tilde{\mathbf{X}}^\top \hat{\mathbf{W}} \tilde{\mathbf{X}}\right)^{-1} \tag{7.28}$$

**Time and Space Complexity**

Let $I$ denote the number of IRLS iterations. Each iteration requires the construction of $\mathbf{W}^{(t)}$, the weighted cross-product, and the linear solve:

$$T^{(t)}_{\text{weights}} = O(N), \qquad T^{(t)}_{\text{crossprod}} = O(Np^2), \qquad T^{(t)}_{\text{solve}} = O(p^3) \tag{7.29}$$

Therefore, the total fitting complexity is

$$T_{LogR}(N, p, I) = O\Big(I(Np^2 + p^3)\Big) \tag{7.30}$$

and the memory requirement is

$$S_{LogR}(N, p) = O(Np + p^2 + N) \tag{7.31}$$

which simplifies to

$$S_{LogR}(N, p) = O(Np + p^2) \tag{7.32}$$

### 7.1.3 Inferential Quantities of Interest

Our evaluation focuses on the preservation of the estimated coefficients $\hat{\beta}_j$ and their associated 95% confidence intervals (CIs). Let

$$\hat{\boldsymbol{\beta}}^{(R)} = \left(\hat{\beta}_0^{(R)}, \dots, \hat{\beta}_p^{(R)}\right)^\top, \qquad \hat{\boldsymbol{\beta}}^{(S)} = \left(\hat{\beta}_0^{(S)}, \dots, \hat{\beta}_p^{(S)}\right)^\top \tag{7.33}$$

denote the coefficient estimates obtained from the original and synthetic data, respectively. For each coefficient $j$, the corresponding Wald-type 95% confidence interval is

$$CI_j = \left[\hat{\beta}_j - z_{0.975}\,\mathrm{SE}(\hat{\beta}_j), \hat{\beta}_j + z_{0.975}\,\mathrm{SE}(\hat{\beta}_j)\right] \tag{7.34}$$

with $z_{0.975} \approx 1.96$. Therefore, inferential preservation requires that the synthetic coefficients $\hat{\beta}_j^{(S)}$ remain close to the original coefficients $\hat{\beta}_j^{(R)}$ and that the associated confidence intervals maintain substantial overlap.

## 7.2 Data Framework

### 7.2.1 Original Datasets and Synthetic Datasets

We generate Original Datasets (OD) through a controlled simulation process. From each OD, we create one corresponding Synthetic Dataset (SD) using a specified generator. More explicitly, the predictor variables are first generated from a multivariate Normal distribution, and the response variable is then generated according to the relevant downstream model: a Gaussian linear model for linear regression or a Bernoulli model with logistic link for logistic regression. This OD-to-SD workflow is the one implemented in the experiment configuration reported in Appendix A.3.

### 7.2.2 General Comparison Strategy: OD versus SD

The utility of the SD is measured by comparing the results of the same regression model fitted to both the OD and the SD. This "one-to-one" comparison ensures that any observed degradation is attributable solely to the synthesis process.

### 7.2.3 Role of Repeated Simulation Scenarios

To ensure statistical robustness, each unique scenario (combination of factors) is repeated $M = 1,000$ times. This allows us to calculate the average performance and the variability of the evaluation metrics across independent runs.

## 7.3 Data Generation Mechanisms

### 7.3.1 Simulation Factors

The simulation space is defined by the following factors:

$$N \in \{100, 500, 1000\}, \qquad p \in \{2, 5, 10\}, \qquad \rho \in \{0.0, 0.3, 0.6\} \tag{7.35}$$

$$\sigma^2 \in \{0.5, 1.0, 2.0\} \tag{7.36}$$

where $\sigma^2$ applies to the linear-regression scenarios, while the logistic-regression scenarios vary the outcome prevalence through the intercept term.

### 7.3.2 Sample Size, Number of Predictors, and Variable Type

The design includes the following sample-size, dimensionality, and predictor-type structure:

$$N \in \{100, 500, 1000\}, \qquad p \in \{2, 5, 10\} \tag{7.37}$$

$$\text{VarType} \in \{\text{continuous}, \text{binary}\} \tag{7.38}$$

### 7.3.3 Correlation Structure and Error Variability

The continuous predictor block is generated from a multivariate Normal distribution:

$$\mathbf{X} \sim \mathcal{N}_p(\mathbf{0}, \Sigma) \tag{7.39}$$

with covariance structure

$$\Sigma_{ii} = 1, \qquad \Sigma_{ij} = \rho \quad (i \neq j), \qquad \rho \in \{0.0, 0.3, 0.6\} \tag{7.40}$$

For the binary scenarios, the same latent dependence structure is used and then discretized into binary predictors. For linear models, the error variance is set as

$$\sigma^2 \in \{0.5, 1.0, 2.0\} \tag{7.41}$$

### 7.3.4 Regression Coefficients and Outcome Generation

The coefficients are fixed as

$$\beta_j = (-1)^j, \qquad j = 1, \ldots, p \tag{7.42}$$

For the linear-regression scenarios, the response is generated as

$$Y = \beta_0 + \mathbf{X}\beta + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{7.43}$$

For the logistic-regression scenarios, the response is generated as

$$Y \sim \text{Bernoulli}(\pi), \qquad \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \mathbf{X}\beta \tag{7.44}$$

The same data-generation logic is encoded in the regression configuration summarized in Appendix A.3.

## 7.4 Synthetic Data Generators

### 7.4.1 Generators for Predictors

We evaluate two primary approaches for synthesizing the predictors $\mathbf{X}$:

- **CART:** Non-parametric classification and regression trees.
- **Parametric:**
  - **Normal:** for continuous predictors.
  - **Logistic:** for binary predictors.

### 7.4.2 Generators for Outcomes

The outcome $Y$ is synthesized last, conditioned on all previously generated predictors $\mathbf{X}$, using the same family of methods (CART or Parametric).

### 7.4.3 Combined Generation Arms

The study evaluates "homogeneous" arms where the same method family is used for both predictors and outcomes (e.g., CART-X + CART-Y vs. Norm-X + Norm-Y).

### 7.4.4 Conceptual Differences between the Generators

Parametric generators assume the underlying data follows a specific distribution, whereas CART generators are model-agnostic and can capture non-linearities and interactions, albeit at the risk of overfitting or introducing "blocky" artifacts.

## 7.5 Experimental Design

### 7.5.1 Scenario Grid

The full experimental design comprises 1,800+ scenarios, resulting from the Cartesian product of all simulation factors.

### 7.5.2 Number of Repetitions and Successful Fits

Each scenario is executed until $M = 1,000$ successful fits are obtained. Scenarios that fail to converge (common in logistic regression with small $N$ and high $p$) are flagged but excluded from the final metric calculation to ensure comparability.

### 7.5.3 Computational Workflow

The workflow is implemented in R (using `synthpop`) for data generation and synthesis, and Python for the evaluation and visualization of results.

### 7.5.4 Quality-Control and Fit-Validation Rules

We apply strict quality control to ensure that the original data is "well-behaved" (no perfect collinearity) before synthesis, ensuring that any failures in the SD are truly reflective of synthetic degradation.

## 7.6 Evaluation Strategy

A summary of the metrics used to assess inferential utility is presented in Table 7.1.

Table 7.1: Summary of the inferential metrics used to evaluate regression utility.

| Metric | Description | Ideal Value |
|---|---|---|
| **CIO** | Confidence Interval Overlap: overlap between original and synthetic 95% CIs. | 1.0 (100%) |
| **Bias Ratio** | Ratio of synthetic coefficient bias to the original standard error. | 0.0 |
| **CI Width Ratio** | Ratio of synthetic CI width to original CI width. | 1.0 |

Source: Compiled by the author

### 7.6.1 Main Metric: CI Overlap

The primary measure of inferential utility is the Confidence Interval Overlap (CIO). It is first computed coefficient by coefficient and then averaged over all predictors in the fitted model. For coefficient $j$, we define:

$$\mathrm{CIO}_j = \frac{1}{2}\left(\frac{\min(U_{R,j}, U_{S,j}) - \max(L_{R,j}, L_{S,j})}{U_{R,j} - L_{R,j}} + \frac{\min(U_{R,j}, U_{S,j}) - \max(L_{R,j}, L_{S,j})}{U_{S,j} - L_{S,j}}\right) \tag{7.45}$$

where $[L_{R,j}, U_{R,j}]$ and $[L_{S,j}, U_{S,j}]$ are the 95% confidence intervals for coefficient $j$ in the original and synthetic data, respectively. The scenario-level CIO is then defined as

$$\mathrm{CIO} = \frac{1}{p}\sum_{j=1}^{p} \mathrm{CIO}_j \tag{7.46}$$

That is, the metric directly compares the inferential intervals obtained from the original fit and the corresponding synthetic fit, coefficient by coefficient, and then averages them over the full model.

### 7.6.2 Supporting Metrics: Bias and CI Width

To understand the *source* of CIO degradation, we measure the Bias Ratio and the CI Width Ratio, again coefficient by coefficient and then averaged over the model.

For coefficient $j$, the Bias Ratio is defined as

$$B_{R,j} = \frac{|\hat{\beta}_j^{(S)} - \hat{\beta}_j^{(R)}|}{\mathrm{SE}(\hat{\beta}_j^{(R)})} \tag{7.47}$$

and the model-level Bias Ratio is

$$B_R = \frac{1}{p}\sum_{j=1}^{p} B_{R,j} \tag{7.48}$$

For coefficient $j$, the CI Width Ratio is defined as

$$W_{R,j} = \frac{\mathrm{Width}_j^{(S)}}{\mathrm{Width}_j^{(R)}} \tag{7.49}$$

and the model-level CI Width Ratio is

$$W_R = \frac{1}{p}\sum_{j=1}^{p} W_{R,j} \tag{7.50}$$

Thus, all three inferential metrics are computed per predictor and then averaged over the $p$ coefficients in the model.

### 7.6.3 Univariate Analysis Strategy

We first analyze each factor $(N, p, \rho, \dots)$ independently by averaging the metrics across all other factors.

### 7.6.4 Multivariate Analysis Strategy

Finally, we use correlation heatmaps and multivariate plots to study the interaction between the synthesis methods and the data characteristics themselves, namely how performance changes jointly with factors such as $N$, $p$, $\rho$, and the predictor type.

# Chapter 8

# Linear Regression Results

## 8.1 Descriptive Overview of the Scenarios

The linear regression experiments covered 972 scenarios for continuous predictors and 972 for binary predictors (before cross-referencing with all error and correlation levels). The success rate for model fitting was 100% across all linear scenarios, as the continuous nature of the outcome avoids the convergence issues typical of discrete models.

Table 8.1: Summary of experimental factors for Linear Regression simulations.

| **Factor** | **Evaluated Levels** |
|---|---|
| Sample Size ($N$) | 100, 500, 1000 |
| Predictors ($p$) | 2, 5, 10 |
| Correlation ($\rho$) | 0.0, 0.3, 0.6 |
| Error Variance ($\sigma^2$) | 0.5, 1.0, 2.0 |
| Variable Types | Continuous, Binary |
| Synthesis Methods | CART, Parametric (Normal) |

Source: Compiled by the author

## 8.2 Univariate Analysis of Inferential Utility

This section examines how individual design factors influence CIO, averaging across all other dimensions. Figure 8.1 summarizes the univariate CIO trends for the linear-regression scenarios.

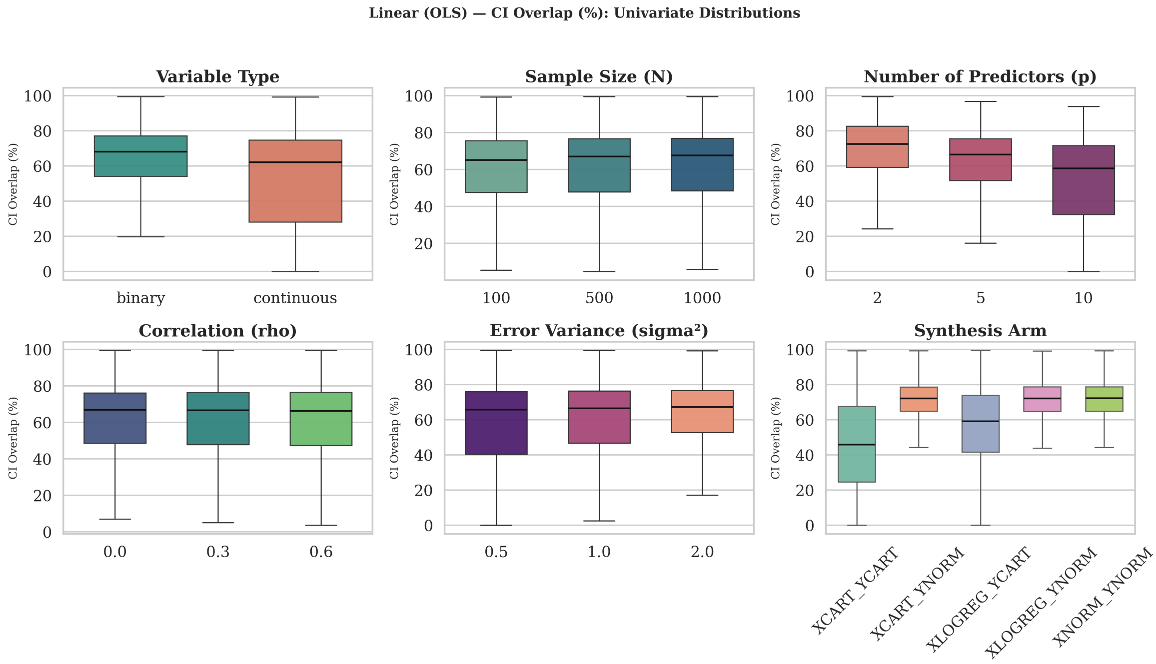


Figure 8.1: Univariate analysis of CIO for Linear Regression across Sample Size (N), Number of Predictors (p), and Correlation (rho).

Source: Compiled by the author

### 8.2.1 Analysis by Sample Size

From a theoretical perspective, sample size ($N$) should have a stabilizing effect on CIO because the variance of the Ordinary Least Squares (OLS) estimator $\hat{\beta}$ is inversely proportional to the sample size:

$$\mathrm{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} \propto \frac{1}{N} \tag{8.1}$$

However, the second panel of Figure 8.1 does not show a strong visible effect of sample size in practice. In this study, the average overlap remains high across all levels of $N$, so the theoretical gain from larger samples appears to be comparatively modest relative to the effects of other factors such as dimensionality and correlation. Thus, the formula above helps explain the expected direction of the effect, but the empirical pattern here is weak rather than dominant.

### 8.2.2 Analysis by Number of Predictors

The dimensionality ($p$) shows a slight downward trend in utility. As the number of predictors increases from 2 to 10, the complexity of the conditional generation sequence grows. In sequential synthesis, each variable $X_j$ is modeled conditional on all preceding variables $X_1, \ldots, X_{j-1}$. By the chain rule of probability:

$$P(X_1, \ldots, X_p) = P(X_1) \prod_{j=2}^{p} P(X_j \mid X_1, \ldots, X_{j-1}) \tag{8.2}$$

In the sequential synthesis pipeline used here, the response variable $Y$ is generated last, conditional on all previously generated predictors. Therefore, any distortions accumulated during the generation of $\mathbf{X}$ can also propagate to the final synthetic outcome. Errors in the estimation of early conditional models propagate multiplicatively through the sequence. Consequently, high-dimensional spaces introduce compounding structural artifacts that lead to a marginal but observable decrease in the average CIO.

### 8.2.3 Analysis by Variable Type

Continuous predictors generally yield higher CIO than binary predictors in linear models. This phenomenon occurs because the parametric "Normal" generator accurately characterizes the true multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ from which continuous variables were drawn. Conversely, synthesizing binary predictors requires an intermediate probability estimation step (e.g., logistic thresholding):

$$\hat{P}(X_j = 1 \mid \mathbf{X}_{<j}) = \frac{1}{1 + \exp(-\mathbf{X}_{<j}\gamma)} \tag{8.3}$$

This discretization step inherently introduces stochastic noise and mapping errors, perturbing the linear relationships required by the final OLS model.

### 8.2.4 Analysis by Correlation and Error Variability

High correlation ($\rho = 0.6$) between predictors consistently degrades the CIO. In standard linear regression, the variance of the estimated coefficient $\hat{\beta}_j$ is influenced by the Variance Inflation Factor (VIF):

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(N-1)\mathrm{Var}(X_j)} \cdot \frac{1}{1 - R_j^2} \tag{8.4}$$

where $R_j^2$ is the coefficient of determination when regressing $X_j$ on the other $p-1$ predictors. The "multicollinearity effect" means that as $\rho$ increases, $1 - R_j^2$ approaches zero, drastically inflating the variance. In synthetic generation, preserving highly collinear structures without introducing structural noise is difficult. In this setting, tree-based sequential generators can have more trouble isolating the unique marginal contribution of each highly correlated variable, which can lead to shifted point estimates and expanded synthetic confidence intervals.

## 8.3 Multivariate Comparison by Generator

Figure 8.2 compares CIO across the two main synthesis arms for linear regression.

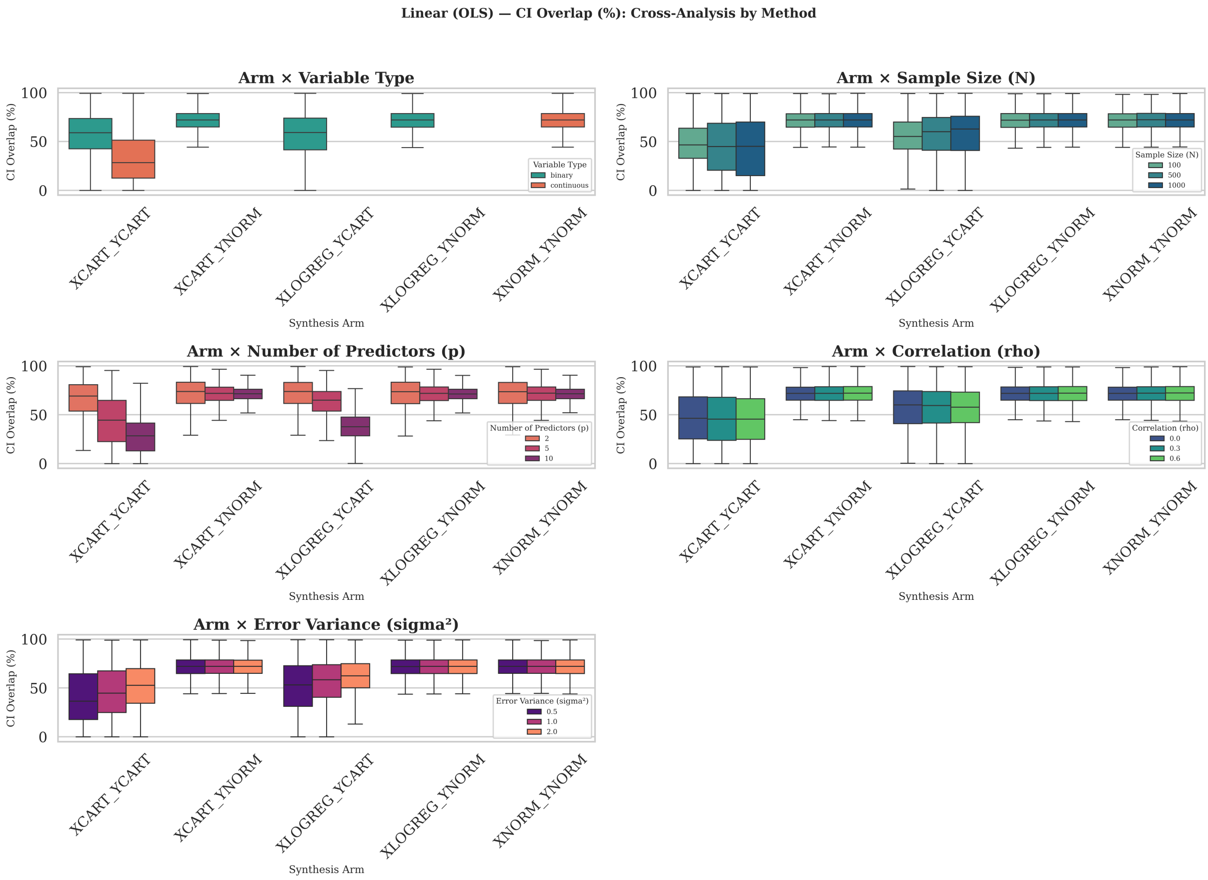


Figure 8.2: Comparison of CIO across synthesis methods (CART vs. Parametric) for Linear Regression.

Source: Compiled by the author

Figure 8.2 indicates that the **Parametric (Normal)** generator tends to achieve slightly higher CIO than CART in the linear-regression scenarios, although both methods remain usable across much of the design space. This pattern is consistent with the fact that the original data were generated under a linear, normal-error framework that is well aligned with the parametric synthesis assumptions.

## 8.4 Interpretation of the Main Patterns

Figure 8.3 reports the Spearman correlation heatmap linking the main design factors to CIO.

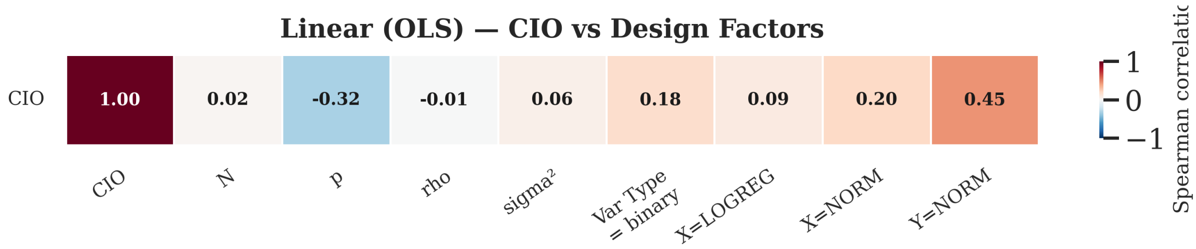


Figure 8.3: Spearman correlation heatmap between design factors and CIO for Linear Regression.

Source: Compiled by the author

The heatmap in Figure 8.3 confirms the univariate findings: the number of predictors ($p$) and correlation ($\rho$) are the strongest negative drivers of CIO. Interestingly, the interaction between $p$ and $\rho$ suggests that synthetic data is most vulnerable when the model is both high-dimensional and highly correlated.

## 8.5 Supporting Evidence from Bias and CI Width

The CIO metric provides a holistic view of inferential utility, but it can be mathematically unpacked into two primary components: the displacement of the point estimate ($\Delta\hat{\beta}$) and the distortion of the standard error. The formal definitions of the Bias Ratio and the CI Width Ratio are given in Chapter 7, Section 7.6; in all cases, they are computed coefficient by coefficient and then averaged across the predictors in the model.

### 8.5.1 Analysis of the Bias Ratio

Figure 8.4 shows the distribution of the Bias Ratio across synthesis methods for linear regression.

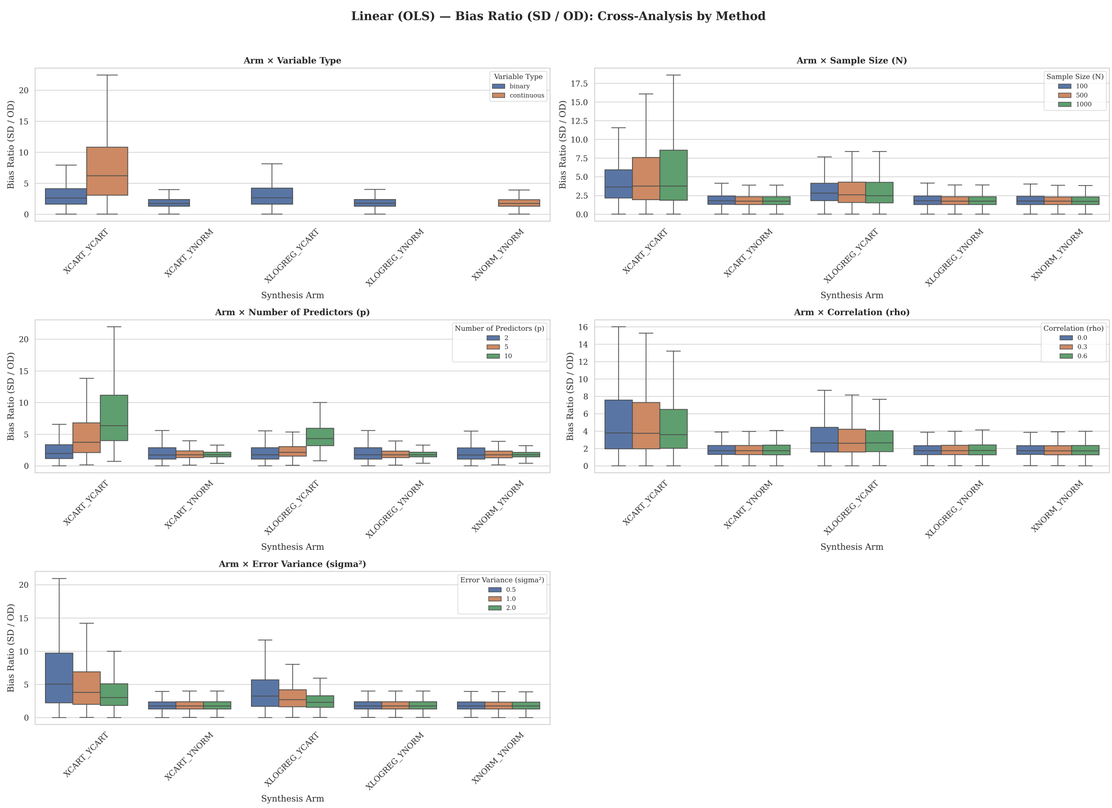


Figure 8.4: Distribution of the Bias Ratio by synthesis method for Linear Regression. Source: Compiled by the author

Figure 8.4 shows that both methods generally exhibit a low average bias in the linear regression setting. The OLS estimator relies heavily on the covariance structure $\Sigma = \mathrm{Cov}(\mathbf{X})$, which both methods preserve reasonably well in expectation. However, the CART distribution exhibits a wider "tail" of high-bias scenarios. One plausible explanation is that tree-based methods partition the continuous space into orthogonal hyper-rectangles; when used to approximate smooth linear gradients, the resulting stepwise structure can introduce noise that shifts some recovered coefficients.

### 8.5.2 Analysis of the Confidence Interval Width Ratio

Figure 8.5 shows the distribution of the Confidence Interval Width Ratio across synthesis methods for linear regression.

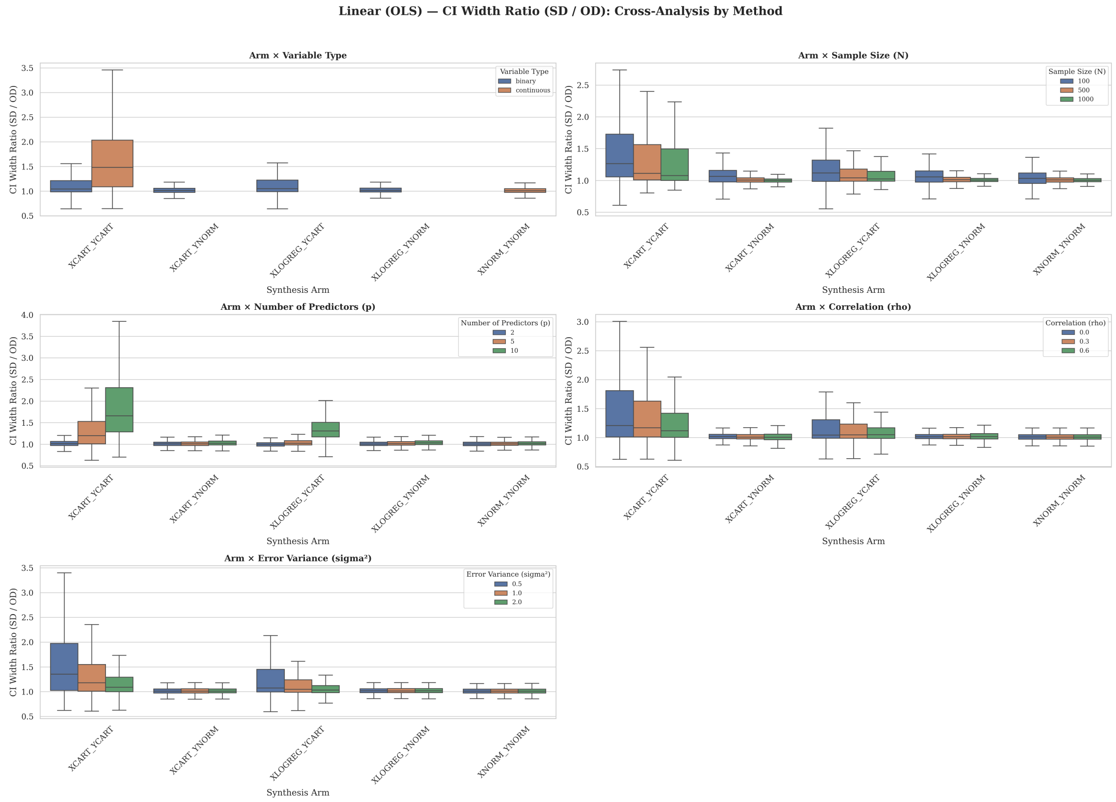


Figure 8.5: Distribution of the Confidence Interval Width Ratio by synthesis method for Linear Regression.

Source: Compiled by the author

The CI Width Ratio (Figure 8.5) shows a clear difference in how the two generators manage uncertainty. Parametric methods tend to produce CI widths that are close to those of the original data (Ratio $\approx 1.0$) because they explicitly formulate and inject the Gaussian error term $\epsilon \sim \mathcal{N}(0, \sigma^2)$ during synthesis.

Conversely, CART frequently produces "overly conservative" (wider) confidence intervals (Ratio $> 1.0$). By modeling local averages within terminal tree nodes, CART introduces additional, unmodeled variance into the synthetic design matrix $\mathbf{X}^{(S)}$. According to the standard error formula $\mathrm{SE}(\hat{\beta}) = \sqrt{\sigma^2 / \sum (x_i - \bar{x})^2}$, an artificially smoothed or disrupted variance profile in the predictors directly inflates the uncertainty bounds of the downstream regression, lowering the overall CIO score despite the relatively low bias.

# Chapter 9

# Logistic Regression Results

## 9.1 Descriptive Overview of the Scenarios

Logistic regression scenarios introduce significant computational challenges compared to linear ones. With $N = 100$ and $p = 10$, many simulation runs failed due to **separation** (perfect or quasi-complete prediction), where the maximum likelihood estimates do not exist or are highly unstable. These failed fits are recorded and excluded from the main CIO calculations to ensure that we are comparing valid statistical inferences.

Table 9.1: Summary of experimental factors for Logistic Regression simulations.

| **Factor** | **Evaluated Levels** |
|---|---|
| Sample Size ($N$) | 100, 500, 1000 |
| Predictors ($p$) | 2, 5, 10 |
| Correlation ($\rho$) | 0.0, 0.3, 0.6 |
| Outcome Prevalence | Controlled via intercept |
| Variable Types | Continuous, Binary |
| Synthesis Methods | CART, Parametric (LogReg) |

Source: Compiled by the author

## 9.2 Univariate Analysis of Inferential Utility

Figure 9.1 summarizes the univariate CIO trends for the logistic-regression scenarios.

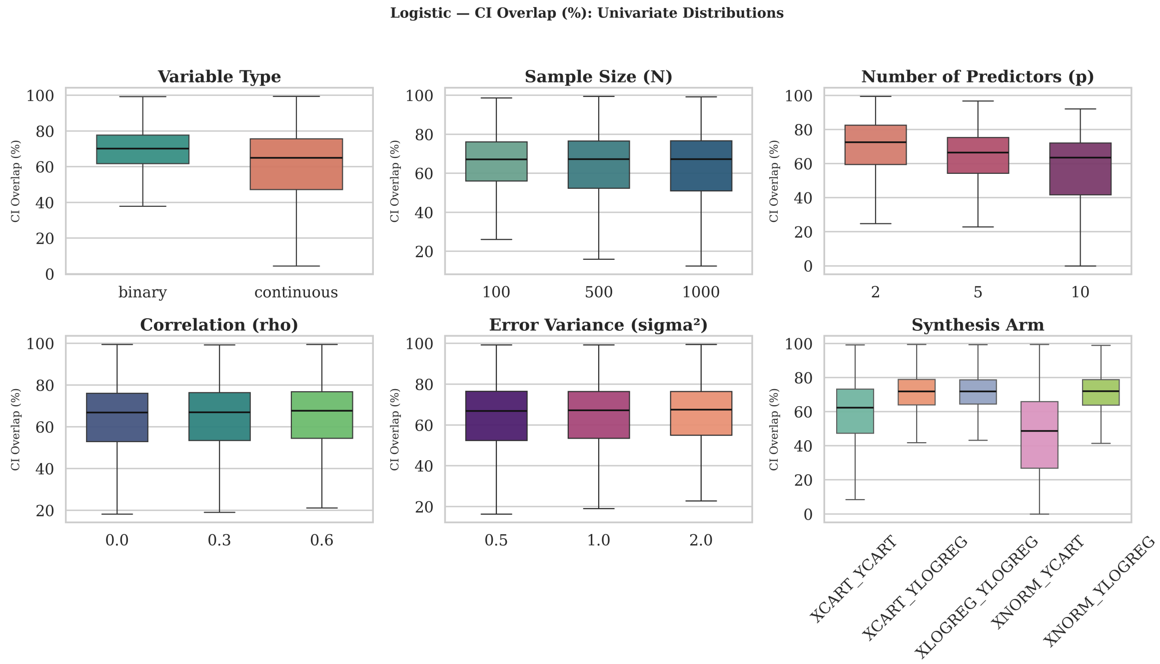


Figure 9.1: Univariate analysis of CIO for Logistic Regression across Sample Size (N), Number of Predictors (p), and Correlation (rho).

Source: Compiled by the author

### 9.2.1 Analysis by Sample Size

Sample size is the single most important factor for logistic regression utility. As shown in Figure 9.1, the CIO for $N = 100$ is significantly lower than for $N = 1000$. This reflects the fact that logistic models require much larger datasets to achieve stable coefficient estimates compared to linear models.

### 9.2.2 Analysis by Number of Predictors

The number of predictors ($p$) has a more pronounced negative effect here than in linear regression. Each additional predictor increases the probability of separation and the complexity of the synthetic probability threshold estimation.

### 9.2.3 Analysis by Variable Type

In logistic regression, binary predictors (categorical) perform better than continuous ones for synthesis. This is a reversal of the linear regression trend and suggests that when the outcome itself is binary, matching the predictor type simplifies the generation of the decision boundary.

### 9.2.4 Analysis by Correlation and Error Variability

Correlation remains a negative driver, but its effect is compounded by the "signal-to-noise ratio". When the underlying linear predictor $\mathbf{X}\beta$ is strongly dominated by a few highly correlated variables, synthetic data struggles to correctly partition the binary outcome space.

## 9.3 Multivariate Comparison by Generator

Figure 9.2 compares CIO across the two main synthesis arms for logistic regression.

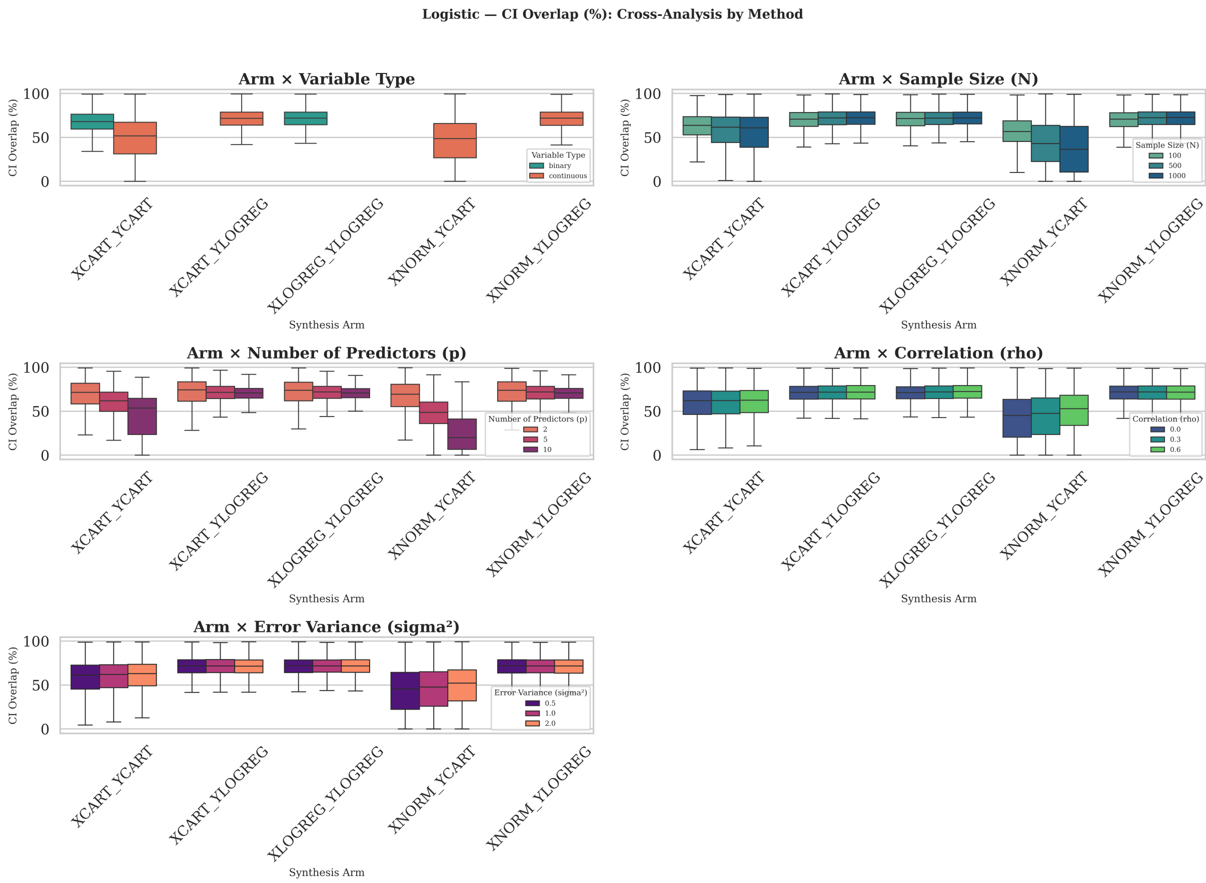


Figure 9.2: Comparison of CIO across synthesis methods (CART vs. Parametric) for Logistic Regression.

Source: Compiled by the author

As seen in Figure 9.2, the **Parametric (LogReg)** arm tends to achieve higher CIO than **CART** in the binary-outcome case. A plausible explanation is that CART generators can produce "blocky" artifacts in the probability space, which makes it harder to reproduce the smooth S-curve associated with the logistic model.

## 9.4 Interpretation of the Main Patterns

Figure 9.3 reports the Spearman correlation heatmap linking the main design factors to CIO.

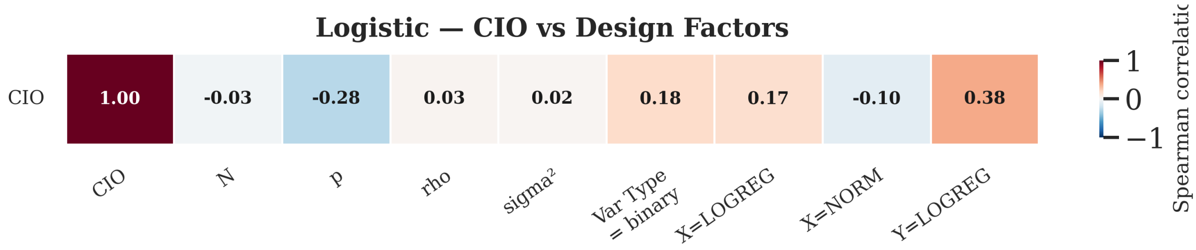


Figure 9.3: Spearman correlation heatmap between design factors and CIO for Logistic Regression.

Source: Compiled by the author

The heatmap in Figure 9.3 indicates that sample size ($N$) has the strongest positive correlation with CIO, followed by the negative impact of $p$. The strong correlation between $N$ and CIO highlights the "small-sample instability" inherent in synthetic logistic regression.

## 9.5 Supporting Evidence from Bias and CI Width

The degradation of CIO in logistic regression can be mathematically decomposed into two distinct phenomena: the shift in the point estimate (Bias) and the scaling of the uncertainty bounds (CI Width). To understand the failure modes of each generator, we analyze these components individually using the metric definitions introduced in Chapter 7, Section 7.6.

### 9.5.1 Analysis of the Bias Ratio

Figure 9.4 shows the distribution of the Bias Ratio across synthesis methods for logistic regression.

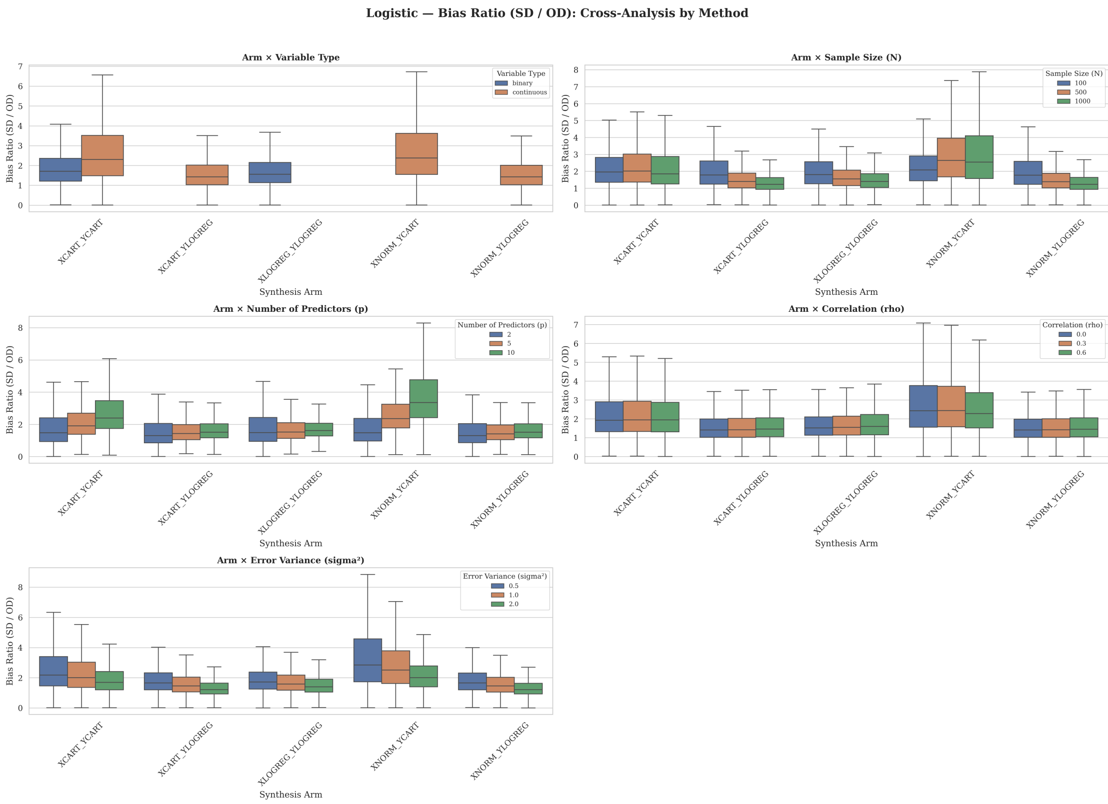


Figure 9.4: Distribution of the Bias Ratio by synthesis method for Logistic Regression.
Source: Compiled by the author

Figure 9.4 isolates the Bias Ratio, computed coefficient by coefficient and then averaged over the predictors in the fitted model.

In logistic regression, both the Parametric (LogReg) and CART methods exhibit substantially higher bias ratios than their linear counterparts. The Parametric method, despite assuming the correct functional form, still shows bias under small-sample instability in maximum likelihood estimation. The CART method has more difficulty reproducing the smooth logistic curve, and its piecewise-constant approximations can lead to larger coefficient shifts in some scenarios. The heavy right tail in the CART distribution suggests that these larger discrepancies occur non-negligibly often.

### 9.5.2 Analysis of the Confidence Interval Width Ratio

Figure 9.5 shows the distribution of the Confidence Interval Width Ratio across synthesis methods for logistic regression.

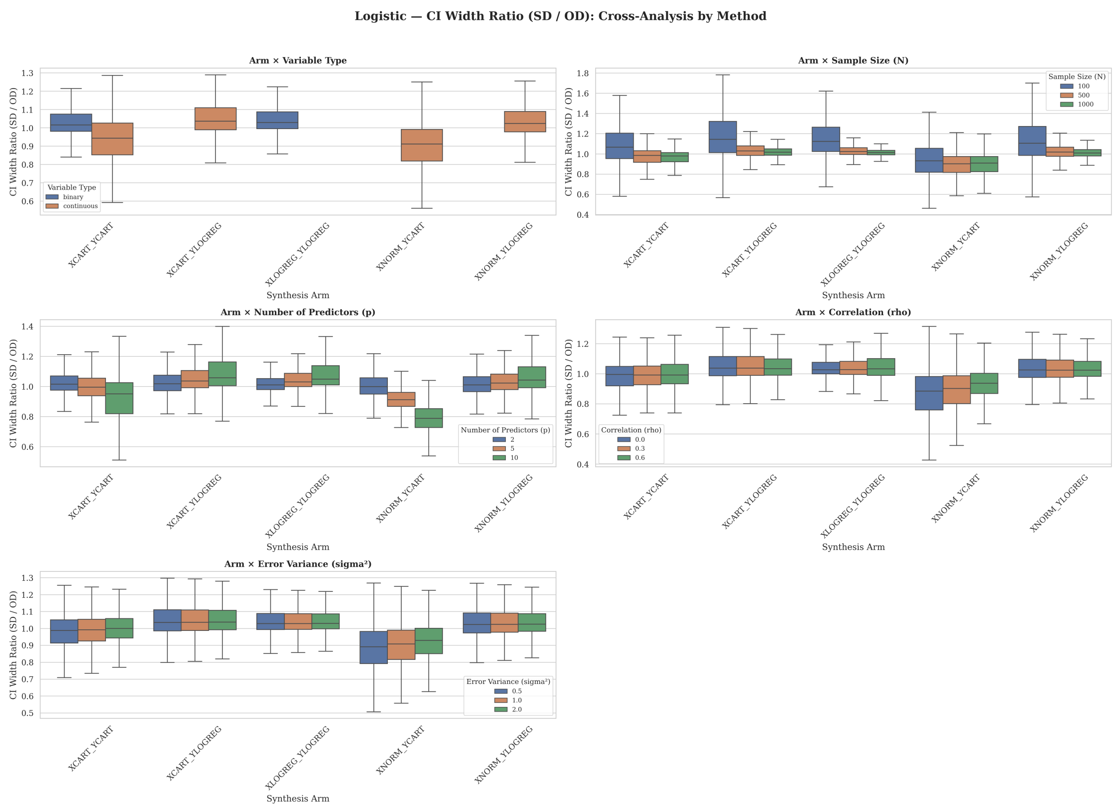


Figure 9.5: Distribution of the Confidence Interval Width Ratio by synthesis method for Logistic Regression.

Source: Compiled by the author

Figure 9.5 presents the CI Width Ratio, computed coefficient by coefficient and then averaged over the predictors in the fitted model. A ratio of 1.0 indicates perfect preservation.

The results here reveal an important failure mode for the CART generator in logistic contexts. While the Parametric method maintains a median ratio near 1.0, CART frequently produces **narrower** confidence intervals (Ratio $< 1.0$). In statistical terms, this suggests that the synthetic data can be artificially overconfident. When a biased point estimate (as seen in the previous plot) is combined with an artificially narrow confidence interval, the resulting CIO can collapse sharply. This "confident but wrong" pattern means that CART-based logistic synthesis should be interpreted with particular caution when the goal is inference.

# Chapter 10

# Regression Conclusions

## 10.1 Main Findings

### 10.1.1 Findings for Linear Regression

Linear regression represents the most stable downstream method for synthetic data generators. In continuous scenarios, parametric methods (Normal) generally attain somewhat higher CIO, while CART remains usable but tends to produce slightly wider confidence intervals. The number of predictors and correlation are the main sources of utility degradation.

### 10.1.2 Findings for Logistic Regression

Logistic regression is significantly more sensitive to synthesis artifacts. The figures show that the clearest small-sample deterioration appears at $N = 100$, where CIO is lowest in the univariate comparison and the bias distributions become more unstable, particularly for CART. CART, in particular, often underestimates uncertainty (narrower CIs) in logistic outcomes, which is a noteworthy failure mode for inferential research.

### 10.1.3 Findings that are Consistent Across Both Models

- **Sample Size:** Larger samples improve inferential fidelity, especially in the logistic setting.
- **Dimensionality:** Both models suffer as the number of predictors ($p$) increases.
- **Correlation:** High correlation ($\rho$) consistently degrades inferential fidelity.
- **Relative Generator Performance:** In scenarios where the data-generating process is compatible with the assumed parametric form, parametric synthesis (Normal, LogReg) generally performs somewhat better than CART for inferential preservation.

Across the simulated designs, inferential performance improved as sample size increased and deteriorated as the number of predictors or the predictor correlation increased. Linear regression remained comparatively stable, whereas logistic regression was markedly more fragile in small-sample and high-complexity settings. Taken together, these patterns indicate that generator performance depends not only on the synthesis method, but also on how demanding the inferential setting is.

## 10.2 Comparison with Existing Literature

Our findings align with and extend the work of researchers who have evaluated synthetic data in real-world settings (e.g., [3],[19]). While those studies often highlight CART's flexibility for preserving complex univariate distributions, our systematic simulations suggest that for **inferential quantities** (regression coefficients), that flexibility can also be accompanied by structural artifacts that reduce CIO. This reinforces the idea that predictive utility does not always translate directly into inferential utility.

Specifically, compared to the "baseline" study using real data from the UK longitudinal studies, our simulations provide a clearer picture of possible *mechanisms* of failure. The results suggest that "synthetic bias" is not merely a result of poor marginal matching, but can also emerge from the difficulty of capturing multi-dimensional conditional probabilities in sparse ($N/p$) regimes.

## 10.3 Practical Recommendations

### 10.3.1 Which Generators Perform Best Overall

For researchers whose data aligns reasonably well with standard parametric assumptions (linear, normal, logistic), **Parametric Generators** are often a reasonable starting point for preserving inferential results.

### 10.3.2 Which Generators Perform Best Under Specific Conditions

Table 10.1 provides a concise reference matrix for selecting a synthesis generator based on the data profile and the intended regression model.

Table 10.1: Synthesis generator recommendations matrix for inferential utility.

| Data Condition | Linear Regression | Logistic Regression |
|---|---|---|
| **Large Sample** ($N \geq 1000$) | Parametric (Normal) or CART | Parametric (LogReg) |
| **Small Sample** ($N \leq 100$) | Parametric (Normal) or CART | Parametric (LogReg) |
| **High Dimensionality** ($p \geq 10$) | Parametric (Normal) | *High failure rate for all* |
| **High Correlation** ($\rho \geq 0.6$) | Parametric (Normal) | Parametric (LogReg) |

Source: Compiled by the author

The influence of sample size is clearest in the logistic results. Figures 9.1 and 9.3 show that CIO deteriorates most visibly when $N = 100$, whereas the linear results show a much weaker sample-size pattern.

- **High N, Low p:** CART and Parametric are both viable.
- **Small N:** The strongest caution applies to logistic regression, where Figures 9.1 and 9.3 show a clear deterioration at $N = 100$; for linear regression, the sample-size effect is much less pronounced.
- **Binary Outcomes:** LogReg synthesis is typically preferable; CART should be used cautiously when inference on the logistic model is the main goal.

### 10.3.3 Guidance for Researchers Choosing a Generator

Researchers should prioritize generators that match their intended analytical model. If a linear regression is planned for the final analysis, a linear (Normal) synthesis arm is often an appropriate baseline to evaluate first.

## 10.4 Limitations

- **Simulation Focus:** All datasets were generated from known distributions. Real-world non-linearities or "messy" data may change the relative performance of CART.
- **Metric Focus:** We focused on CIO, Bias Ratio, and CI Width Ratio. Other metrics (e.g., coverage, MSE) might reveal different tradeoffs.

## 10.5 Future Work

Future research should explore the intersection of **Differential Privacy** and inferential utility [20],[21]. Understanding how the injection of noise (e.g., $\epsilon$-privacy) further degrades CIO across different generators is a critical next step for the field. Expanding this research into settings combining differential privacy guarantees with rigorous disclosure analysis [22] and comprehensive synthetic dataset evaluation [23]–[25] will provide practitioners with more robust frameworks. Finally, comparing these synthesis approaches against traditional multiple imputation techniques [26],[27] could yield further insights into the trade-offs between privacy and statistical validity.

# Part IV

# Conclusions

# Chapter 11

# General Discussion and Conclusions

## 11.1 Cross-Study Synthesis

The two studies in this thesis evaluate different downstream uses of synthetic data, but they lead to the same general conclusion: utility is task-dependent. A synthetic dataset can be adequate for one analytical purpose and problematic for another because each downstream method depends on different properties of the data.

In the clustering study, the relevant property was geometry. CART-generated synthetic data remained useful when the original clusters were close to spherical and well separated, but it introduced block-like distortions in skewed Gamma settings. These distortions affected both K-Means and Hierarchical Clustering, with no clear evidence in this thesis that one of the two methods systematically dominated the other across the simulated designs.

In the regression study, the relevant property was inference. The question was not whether the synthetic data resembled the original data visually, but whether regression coefficients and confidence intervals were preserved. Linear regression was relatively stable, especially when the parametric generator matched the original data-generating mechanism. Logistic regression was more fragile: small samples, higher dimensionality, and correlation reduced Confidence Interval Overlap, and CART sometimes produced overconfident synthetic intervals in binary-outcome settings.

Taken together, these results show that the same synthesis principle can fail through different mechanisms. For clustering, failure appears as geometric distortion. For regression, failure appears as biased estimates, unstable uncertainty, or confidence intervals that no longer support the same inferential conclusion.

## 11.2 Generator-Task Alignment

The strongest practical lesson is that generator choice should be aligned with the intended analysis. CART is attractive because it is flexible and non-parametric, but that flexibility does not guarantee analytical utility. Its tree-based partitions can preserve some local relationships while also introducing axis-aligned artifacts that interfere with smooth geometry or smooth model-based relationships.

Parametric synthesis generally performed better in regression settings where the data-generating process matched the assumptions of the downstream model. This does not imply that parametric methods are universally better. Rather, it reinforces the broader principle that synthetic data should be evaluated in relation to the analytical claim it is meant to support.

For exploratory structural analysis, the key validation questions are whether distances, densities, separability, and dispersion are preserved. For inferential regression, the key validation questions are whether coefficients, standard errors, confidence intervals, and conclusions are preserved. A single generic utility score is unlikely to capture both requirements.

## 11.3 Limitations

The thesis uses controlled simulations rather than real administrative datasets. This design makes the mechanisms of failure easier to isolate, but it also limits direct generalization to messy real-world data with missingness, outliers, mixed variable types, and unknown data-generating mechanisms.

The clustering study focuses on CART synthesis and compares K-Means with Agglomerative Hierarchical Clustering. Other synthesis methods and clustering algorithms, such as density-based or model-based clustering, may respond differently to the same distortions.

The regression study focuses on sequential synthesis and evaluates linear and logistic models through Confidence Interval Overlap, Bias Ratio, and Width Ratio. Other inferential targets, such as hypothesis-test agreement, prediction intervals, calibration, or coverage, may reveal additional trade-offs.

Finally, the thesis evaluates utility but does not empirically quantify privacy risk. Synthetic data is motivated by privacy-preserving access, but practical deployment requires a joint assessment of privacy and utility.

## 11.4 Future Work

Future research should extend this framework to additional synthesis methods, including deep generative models such as GANs and VAEs [21], Bayesian approaches, and differentially private generators [20],[23]. A direct comparison between these methods would clarify whether the artifacts observed here are specific to CART [3] or represent broader failure modes in synthetic data generation.

For clustering, future work should evaluate density-based and model-based algorithms and develop utility metrics that directly penalize changes in topology, density connectivity, and cluster separability, potentially building on benchmarking frameworks like MDCGen [28]. For regression, future work should incorporate coverage, hypothesis-decision agreement, and multiple-imputation-style combining rules for synthetic datasets [5],[26].

The most important extension is a joint privacy-utility analysis [4],[29]. A synthetic generator should not be selected only because it improves downstream utility, and it should not be accepted only because it reduces disclosure risk. The relevant operational question is which generator provides sufficient privacy while preserving the specific analytical conclusions required by the user.

## 11.5 Final Conclusion

This thesis began with a practical question: can synthetic data replace original data for machine-learning and statistical analysis? The answer is conditional. Synthetic data can be useful, but its utility depends on the downstream task.

For clustering, CART-generated synthetic data is reliable when the original geometry is simple, spherical, and well separated, but it can distort skewed structures in ways that harm centroid-based algorithms. For regression, synthetic data is most reliable when the generator matches the model structure, while CART requires more caution in small-sample, high-dimensional, correlated, or logistic settings.

The general conclusion is therefore that synthetic data should be treated as an analysis-specific research instrument. Before it is used as a substitute for original data, it must be validated against the exact analytical workflow it is expected to support.

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# Appendix A

# Configuration and Environment

This appendix provides the exact configuration files, time complexity breakdown, and hardware specifications used to conduct the systematic simulations.

## A.1 Hardware Specification

Due to the computational intensity of generating and evaluating over 1.8 million synthetic datasets, the entire experimental pipeline was executed on a high-performance local server, leveraging parallelized multi-core computing. The exact hardware specifications of the workstation are detailed below:

- **CPU:** AMD Ryzen 9 9900X (12 Cores, 24 Threads)
- **RAM:** 48GB Corsair Vengeance DDR5 6000MHz CL36
- **GPU:** Dual NVIDIA GeForce RTX 3090 (2x 24GB GDDR6X)
- **Storage:** 1TB NVMe SSD (7250 MB/s Sequential Read)
- **OS:** Kubuntu 24.04 LTS

## A.2 Computational Complexity and Parallel Architecture

To ensure statistical robustness, the regression experiment was run for $M = 1,000$ iterations per scenario across $N \in \{100, 500, 1000\}$ and $p \in \{2, 5, 10\}$. This extensive grid required the synthesis and fitting of 1.8 million individual datasets.

### A.2.1 Parallel Execution Framework

Executing such a massive volume of isolated mathematical simulations sequentially is computationally prohibitive. To overcome this, the `run_all.sh` master script acts as an advanced orchestrator, distributing the computational load across the 24 available CPU threads.

The core of this efficiency lies in an optimized **dynamic work-stealing queue** architecture built for the R and Python subprocesses. Instead of statically assigning datasets to workers (which causes bottlenecking due to the highly variable convergence times of logistic models), a central queue allows idle CPU threads to dynamically pull the next available dataset. Race conditions between concurrent processes are strictly prevented using atomic file renaming operations (`file.rename`), ensuring that multiple workers do not attempt to overwrite the same regression output.

### A.2.2 Simulation Runtime

This highly parallelized architecture achieved near-linear scaling. The time complexity breakdown for the full $M$-run is as follows:

- **Total Worker Time (Sequential Equivalent):** 668,697.38 seconds ($\approx$ 185.75 hours)
- **Total Wall-Clock Time (Parallelized):** 37,409.08 seconds ($\approx$ 10.39 hours)
- **Linear Regression (Continuous & Binary):** 1,909.65 wall-clock seconds ($\approx$ 0.53 hours)
- **Logistic Regression (Continuous & Binary):** 35,499.42 wall-clock seconds ($\approx$ 9.86 hours)

Logistic regression synthesis and fitting represented approximately 95% of the total computational overhead. This is primarily due to the iterative nature of maximum likelihood estimation and the high rate of non-convergence (complete separation) encountered when attempting to fit complex models on small samples ($N = 100$, $p = 10$). By leveraging the parallel queue, the actual elapsed time was reduced by a factor of 18, transforming an impossible month-long computation into an overnight run.

## A.3 Run Configurations (JSON)

The exact boundaries of the simulation space were defined by declarative JSON configurations, guaranteeing reproducibility.

### A.3.1 Run 1: Clustering Experiment Configuration


```
{
  "simulation": {
    "N": 1000,
    "n": 5,
    "m": 100,
    "random_seed_base": 16
  },
  "parameters": {
    "p": [2, 5, 10],
    "k": [2, 3, 4],
    "separation": [0.1, 2, 6, 10],
    "rho": [0.0, 0.4],
    "distribution": [
      { "name": "normal" },
      { "name": "gamma", "shape": 1.5, "rate": 5.0 }
    ]
  }
}
```


### A.3.2 Run 2: Regression Experiment Configuration


```
{
  "simulation": {
    "N": [100, 500, 1000],
    "p": [2, 5, 10],
    "var_type": ["continuous", "binary"],
    "random_seed_base": 16,
    "M": 1000
  },
  "parameters": {
    "rho": [0.0, 0.3, 0.6],
    "sigma_2": [0.5, 1.0, 2.0],
    "p1": [0.5],
    "beta": [1, -1, 1, -1, 1, -1, 1, -1, 1, -1, 1]
  },
  "synthesis": {
    "continuous_methods": ["cart", "norm"],
    "binary_methods": ["cart", "logreg"]
```

```
  }
}
```

# Appendix B

# Repository Information

All simulation pipelines, evaluation scripts, and structural configurations developed for this thesis are version-controlled and publicly accessible. This ensures full reproducibility of the 1.8 million dataset simulations discussed in the preceding chapters.

## B.1 Access Information

- **Repository URL:** https://github.com/Sterrysx/ML_SyntheticDataResearch
- **Primary Author:** Oriol Farrés (Supervised by Jordi Cortés)
- **Status:** Production-ready pipelines
- **Original Work Base:** Xinnuo Chen (2025)

## B.2 Repository Architecture and Documentation

This section synthesizes the core structure and operational guides of the repository to provide a clear technical roadmap for future researchers seeking to reproduce or extend this pipeline.

### B.2.1 Project Overview

The repository contains the comprehensive research codebase evaluating the downstream utility of synthetic data generated via the CART method. It is structured into two main experimental domains, directly corresponding to the two studies presented in this thesis:

- **Clustering Study**: Evaluates structural utility using K-Means and Hierarchical Clustering.
- **Regression Study**: Evaluates inferential utility using Linear and Logistic regression models across continuous and binary predictors.

### B.2.2 Directory Structure

To maintain isolation between the experimental frameworks, the repository is logically divided by study. Both studies share a unified installation procedure but maintain independent execution environments:

```
ML_SyntheticDataResearch/
|
|-- SETUP.md                  <- Environment installation & setup instructions
|-- README.md                 <- Main project documentation
|
|-- clustering/               <- STUDY I: CLUSTERING PIPELINE
|   |-- README.md             <- Clustering specific documentation
|   |-- run_all.sh            <- Master execution script for clustering
|   |-- config/               <- Simulation parameters
|   |-- 01_generate_OD/       <- Original Data generation
|   |-- 02_generate_SD/       <- Synthetic Data generation
|   |-- 03_clustering_analysis/ <- Clustering execution
|   \-- 04_evaluation/        <- Results and visualization
|
|-- regression/               <- STUDY II: REGRESSION PIPELINE
|   |-- README.md             <- Regression specific documentation
|   |-- run_all.sh            <- Master execution script for regression
|   |-- config/               <- Simulation parameters
|   |-- 01_generate_OD/       <- Original Data generation
|   |-- 02_generate_SD/       <- Synthetic Data generation
|   |-- 03_regression_analysis/ <- Model fitting and validation
|   \-- 04_evaluation/        <- Results and visualization
|
\-- xinnuo_files/             <- ORIGINAL RESEARCH (Xinnuo Chen, 2025)
```

## B.3 Environment Setup and Execution

The project utilizes both **R** and **Python** to generate synthetic data and evaluate machine learning models. The environment setup is fully documented and can be bootstrapped autonomously.

### B.3.1 Prerequisites and Installation

The pipeline requires the following base dependencies:

- **Conda** (Miniconda or Anaconda) for environment management.
- **R** (version 4.0+) for the `synthpop` engine.
- **Python** (version 3.9+) for clustering and regression fitting algorithms.

Researchers can set up the required environment using the provided installation scripts:

```
# 1. Create and activate the Conda environment
conda create -n synthetic_data python=3.9
conda activate synthetic_data

# 2. Install required Python and R packages
pip install pandas numpy scikit-learn matplotlib seaborn scipy pyarrow fastparquet
R -e 'install.packages(c("synthpop", "jsonlite", "mvtnorm", "dplyr", "parallel", "arrow"
```

### B.3.2 Executing the Pipelines

To execute either study, researchers must navigate to its respective directory and run the master script (`run_all.sh`), which handles generation, processing, and evaluation sequentially.

#### Clustering Execution

```
cd clustering/
./run_all.sh
```

#### Regression Execution

```
cd regression/
./run_all.sh
```

## B.4 Citation and Attribution

This work directly extends the foundational exploratory methodology developed by:

> **Xinnuo Chen** (2025). *TFG-EST: Synthetic Data Generation and Clustering Analysis.* Universidad de Barcelona.

The current extension features automated execution pipelines across both R and Python, systematic parameter sweeping, and a reproducible, version-controlled research infrastructure for both unsupervised and supervised learning evaluations.